



ASIA PACIFIC COLLEGE

The School of Computing and Information Technologies

Centralized Booking System for Pet Grooming in Makati: Serving Owners and Groomers

Project Documentation Submitted to the Faculty of the
School of Computing and Information Technologies
Asia Pacific College

In In Partial Fulfillment of the Requirements for
Project Management
PROJMAN

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Initiation

Project Charter

Introduction

This Charter formally authorizes the furlink project to develop and implement a web-based pet grooming booking and management system tailored for dog and cat owners, as well as small to medium-sized pet grooming shops.

A project plan will be developed and submitted to the Project Advisers for approval. The project plan will include: a scope statement, schedule, cost estimate, budget, and provisions for scope, resource, schedule, communications, quality, risk, and stakeholder management. All resources will be provided and assigned by the project team, with financial and logistical support from immediate family members serving as foundational sponsors.

Project Scope

The purpose of furlink is to innovate the way pet grooming services are booked and managed by replacing manual processes with a centralized digital platform. This project directly addresses the need for digital solutions in the local pet care market to reduce scheduling conflicts, enhance customer satisfaction, and improve service visibility for small to medium-sized pet grooming enterprises. To achieve these goals, furlink focuses on eliminating double bookings through an AI Booking Assistant to ensure zero incidents during the first three months of release. Another objective is onboarding and verifying at least one service provider. Additionally, the platform will speed up the booking process by achieving 95% AI accuracy in interpreting user requests, while empowering providers with actionable business insights through an analytics dashboard, targeted for completion by August 3, 2026.

The primary deliverable is a comprehensive web-based platform built on React.js and Supabase. This platform will feature a calendar-based scheduling engine, a service directory, and secure payment gateway integration via PayMongo. Furthermore, it will integrate advanced artificial intelligence capabilities, including a Gemini 2.5 Flash-powered AI Booking Assistant for information retrieval and payment process help, an AI Pet Haircut Generator utilizing Pollinations.ai with Hugging Face failover redundancy, for visual grooming references, and an AI-Powered Data Analyst allowing service providers to have actionable insights on their sales and booking data.

The development and operation of Furlink are governed by strict constraints and security boundaries. The cloud-based system requires constant internet connectivity and must account for and mitigate serverless Edge Function cold-start delays, generative image latency, and Gemini AI rate limits during continuous usage. Legally, the platform must strictly comply with the Data Privacy Act of 2012 (RA 10173), ensuring that raw credit card details are never collected or stored, relying entirely on PayMongo for secure financial transactions. Additionally, the integrated AI modules are constrained by strict Row-Level Security (RLS) and Read-Only Database Views to proactively prevent prompt injection, data leakage, and unauthorized cross-provider data access. Ultimately, the project's success will be determined by the timely deployment of all planned release versions (v1.0 through v2.1), formal validation from academic project advisers, and the secure facilitation of bookings between pet owners and service providers.

Project Manager

The Project Manager, Lalainne Andaya (Project Manager & Developer), is hereby authorized to approve task division, make internal team decisions, and coordinate directly with all project stakeholders, including project advisers, and consultants. The Project Manager is responsible for developing the project plan, monitoring the schedule, cost, and scope of the project during implementation, and maintaining control over the project by measuring performance and taking corrective action.

Milestone

The project plan will be executed in accordance with the milestone schedule below. The Project Manager must approve any schedule changes which may impact milestones and communicate them to the advisers and stakeholders.

Milestones	Date
Capstone 2 Kickoff	March 7, 2026
Final Testing	July 6-7, 2026
Release to Service Provider (Training)	July 8, 2026
furlink v1.0 Deployment	July 11, 2026
Enhancement of Core Booking Modules	July 13-17, 2026
Development of AI Booking Assistant	July 20-30, 2026
System and Integration Testing	July 31, 2026 – August 7, 2026
furlink v2.0 Release	August 10, 2026
User Acceptance Testing & Fixes	August 17, 2026 – August 21, 2026

Solution Release furlink v2.1	August 31, 2026
Project Completion	October 1, 2026

Project Budget

The budget for furlink is ₱420,374.26 Foundational sponsorship to be provided by the project team members and their immediate families as a self-funded startup model.

Business Case

Executive Summary

This business case outlines how the Furlink project will address current operational inefficiencies within the local pet grooming industry, the benefits of implementing this digital platform, and the justification for its development.

Currently, the pet grooming market relies heavily on decentralized, manual booking processes such as facebook messaging and SMS, leading to schedule conflicts, miscommunication regarding service expectations, and revenue loss for pet grooming shops. Furlink provides a centralized, web-based booking system equipped with calendar-based scheduling, an AI booking assistant, and business analytics. By transitioning local grooming shops to this platform, furlink will reduce administrative overhead, eliminate double bookings, improve provider visibility, and enhance the overall customer experience for dog and cat owners.

1.1 Issue

Despite rapid growth in the Philippine pet care industry, local grooming service providers lack structured booking systems. Currently, manual booking process causes frequent scheduling conflicts, with 33% of businesses reporting recurring double-booking incidents according to an article by Global Pets Philippines. Furthermore, pet owners experience unmet service expectations due to reliance on verbal instructions without visual references, and local shops suffer from low market visibility and a lack of actionable business data to track their performance.

1.2 Anticipated Outcomes

Deploying Furlink will centralize the booking and management process, allowing service providers to transition away from manual tracking. The anticipated outcomes include:

- **Elimination of Double Bookings:** Automated, real-time calendar synchronization will prevent scheduling overlaps.
- **Improved Service Accuracy:** The AI booking assistant will enhance the pet owner's experience through providing relevant help during the booking process.
- **Enhanced Business Operations:** Service providers will gain access to a dedicated dashboard tracking monthly revenue, customer feedback, and booking trends.
- **Increased Visibility:** Small to medium-sized grooming shops will reach a broader market beyond their immediate physical location through the platform's localized directory.

1.3 Recommendation

It is recommended to proceed with the development and deployment of the Furlink web platform using a modern, cost-effective technology stack (Vercel, Supabase, React.js, and Node.js). This approach allows LogiTeh to deliver a robust Minimum Viable Product (MVP) tailored specifically to the Makati City market. Furlink will methodically onboard partner salons to validate the system in a real-world environment.

1.4 Justification

The Furlink project was selected because it directly addresses the identified pain points of a rapidly expanding market segment while maintaining strict adherence to a student-led startup budget. Relying on the traditional booking practices for pet grooming services guarantees continued revenue loss for small businesses due to operational inefficiencies. Other existing digital alternatives are either geographically restricted (Rover), require expensive monthly subscriptions (DaySmart), or lack upfront pricing transparency (PetBacker). Furlink offers a hyper-localized, transparent, and financially viable solution that aligns with modern digital transformation goals.

Business Case Analysis Team

The following individuals comprise the core project and analysis team responsible for the creation and execution of the Furlink project business case.

Role	Description	Name/Title
Project Manager	Manages the project team, milestones, and backend development.	Lalaine Andaya, Project Manager & Developer

Lead Full stack Developer	Leads software development and system architecture.	Atasha Frances Gayle Doria, Lead Developer
Full stack Developer	Responsible for the core feature development implementation, and maintenance of both the user interface and backend application logic.	Feliz Angelica Salting, Release Manager / Developer
QA Lead	Manages testing, quality assurance, and documentation validation.	Michelle Reina Pineda, QA Lead

Problem Definition

3.1 Problem Statement

Pet owners struggle to find, compare, and reliably book pet grooming services, while local small-to-medium pet grooming shops suffer from inefficient operational processes. Grooming shops currently track bookings manually using notebooks, whiteboards, and social media direct messages, resulting in high rates of missed appointments, double bookings, and long waiting times. Additionally, the lack of standardized assistance when it comes to the booking process leads to the pet owner's dissatisfaction. Finally, independent groomers lack the digital tools necessary to track revenue, gather organized customer feedback, and enforce liability waivers.

3.2 Organizational Impact

For the target grooming shops adopting Furlink, the organizational impact will be significant:

- **Tools:** The transition from physical calendars and separate social media chats to a digital dashboard and centralized booking platform.
- **Processes:** The booking process will shift from manual, asynchronous text negotiations to automated, self-service bookings by pet owners, freeing up staff to focus purely on grooming services rather than admin tasks.
- **Hardware/Software:** Grooming shops will require internet-connected devices (tablets, smartphones, or PCs) to monitor their Furlink provider dashboards and accept/decline bookings.

3.3 Technology Migration

Since target businesses are currently utilizing non-integrated tools such as pen and paper and standalone social media apps, technology migration involves onboarding and digital adoption rather than legacy data transfer.

- Phase I: Shops will create their profiles, input their service listings, set pricing, and define operating hours.
- Phase II: Shops will digitally integrate their terms, conditions, and liability waivers into the platform.
- Phase III: A parallel run where shops may still accept walk-ins or manual messages, but immediately log them into the Furlink calendar to prevent platform conflicts.

Project Overview

4.1 Project Description

Furlink is a web-based booking system that connects pet owners with grooming service providers. It replaces manual scheduling with a calendar-based booking engine. The system integrates advanced features, including an AI booking assistant, and a comprehensive business dashboard for providers to track revenue, completed bookings, and customer feedback.

4.2 Goals and Objectives

This section lists the business goals and objectives which are supported by the project and how the project will address them.

Objective	Description
Ability to Customize Pet Haircut	Launch an AI-powered tool by August 2026 that accurately previews how a haircut will look on a pet with a 70% accuracy based on their specific breed and hair condition.
Reduce Scheduling Conflicts	Integrate an AI Booking Assistant to automate scheduling and eliminate double bookings for providers for at least three months post-launch.
Improve Visibility of Pet Service Providers	Provide a professional platform for local pet care providers, starting with the successful onboarding and verification of at least one service provider within the first three months.
Provide Business Insights	Equip providers with an AI-powered dashboard by August 2026 that automatically tracks monthly sales and

	customer feedback to help them grow their business.
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4.3 Project Performance (KPIs)

The following table lists the key resources, processes, or services and the anticipated business outcomes in measuring the performance of the furlink project. These performance measures will be quantified and further defined in the detailed project plan.

Key Process	Performance Measure
Booking Reliability	Reduce double-booking rate for appointments processed through the platform.
User Adoption	Successful onboarding of initial target shop which is The Groom Room by Steve
AI Booking Assistant Tool Effectiveness	Achieve a successful task completion rate for inquiries and bookings handled with the AI assistant while maintaining accurate context retention

4.4 Project Assumptions

- Target pet grooming service providers have a reliable internet connection and a device to manage their dashboards.
- Pet owners are open to adopting digital solutions for pet care.
- Foundational sponsorship will fully fund the required technology stack (servers, domain, APIs).
- At least one pet grooming service provider will formally onboard for the MVP launch.
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4.5 Project Constraints

- Budgetary: The project is limited to student-led foundational financial resources
- Timeline: Strict adherence to academic Project-Based Learning (PBL) schedules, requiring total project completion by October 2026.

4.6 Major Project Milestones

Milestones	Date
Capstone 2 Kickoff	March 7, 2026
Final Testing	July 6-7, 2026
Release to Service Provider (Training)	July 8, 2026

furlink v1.0 Deployment	July 11, 2026
Enhancement of Core Booking Modules	July 13-17, 2026
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Solution Release furlink v2.1	August 31, 2026
Project Completion	October 1, 2026

Strategic Alignment

The Furlink project aligns directly with the United Nations Sustainable Development Goal (SDG) 8: Decent Work and Economic Growth.

Plan/Goal	Relationship to Project
SDG 8: Economic Growth	By providing a centralized digital platform, Furlink amplifies the local pet service industry, expanding the market reach and economic viability of small-to-medium grooming enterprises that currently suffer from low visibility.
LogiTech Core Vision	Demonstrates the practical application of modern tech stacks to solve real-world, local community operational bottlenecks, fulfilling the academic and entrepreneurial objectives of the team.

Cost Benefit Analysis

The following table captures the cost and savings actions associated with the furlink project, descriptions of these actions, and the financial impact anticipated through the first year of operation. The financial benefits below are calculated based on the operational savings and revenue generation for an initial cohort of one partner grooming shop adopting the system in the first year.

Action	Action Type	Description	First Year Costs (- indicates revenue)
Total Development Budget	Cost	Total upfront investment required to design, build, and deploy the system. This includes labor (₱146,196.99), pre-launch software/APIs (₱21,158.41), and 10% contingency reserve (₱14,619.70).	₱181,975.10
Year 1 Cloud Infrastructure	Cost	Cloud infrastructure, domain registration, and project overhead required for the duration of the project.	₱52,069.40
Year 1 API Usage	Cost	Annual operational cost of running the Gemini AI Booking Assistant and other API integrations at scale.	₱4,380.47
Year 1 Projected Revenue	Revenue	Direct platform revenue generated through combined booking fees and premium ad placements in the first year.	-₱56,590.75
Net First Year Impact		(Total CapEx + Total Year 1 OpEx - Year 1 Revenue)	₱181,834.22

Alternative Analysis

The following alternative options were considered but ultimately not selected for addressing the business problem.

Alternative Option	Reasons For Not Selecting Alternative
No Project (Status Quo)	• Continued high rates of double bookings and scheduling errors. • Lost revenue due to no-shows and lack of enforced liability waivers.
Use Existing Platform: PetBacker	• Unclear business processes; forces users to input detailed pet data before allowing them to explore the platform. • Lack of upfront pricing transparency makes it difficult for owners to compare services within a budget.
Use Existing Platform: Rover	• Geographically restricted; only available and tailored for users in the USA and Canada.
Use Existing Software: DaySmart or Pet Express	• DaySmart: Requires expensive monthly subscription tiers and acts only as an internal tool, offering no public marketplace visibility. • Pet Express: A closed, proprietary system exclusive to Pet Express branches, offering no solution for independent local businesses.

Project Brief

PROJECT NAME	FurLink		
PROJECT MANAGER	Lalaine C. Andaya	EMAIL	lcandaya2@student.apc.edu.ph
START DATE	03/07/2026	BUDGET	₱420,374.26
END DATE	10/1/2026		
PROJECT OVERVIEW	As a startup, furlink aims to modernize the way pet grooming services are booked and managed in the Philippines. Unlike platforms such as Agoda or Airbnb—which focus on property and hotel bookings—		

	furlink specializes in connecting dog or cat owners with pet grooming service providers through a streamlined digital platform. furlink offers a hassle-free experience for both customers and service providers. It enables grooming businesses to advertise their services while allowing pet owners to discover, schedule, and manage appointments in one place. Additionally, the platform integrates an innovative AI-powered grooming preview tool, giving users a visual reference of potential pet haircut styles before booking—bridging the gap between customer expectations and actual grooming outcomes. Designed as a local-first startup solution, furlink initially targets grooming services within select cities, with plans to expand as adoption grows.
OBJECTIVES	The system is designed to achieve four primary goals: 1. Ability to Customize Pet Haircut: Successfully develop and deploy an AI-powered haircut preview tool, integrated into the platform, by August 2026, ensuring it achieves a minimum of 70% image rendering accuracy, accurate identification of selected haircut suitability for the pet's current hair style and condition. 2. Reduce Scheduling Conflicts: Eliminate double booking by ensuring that each service provider does not experience booking incidents during the first three (3) months of the release by integrating an AI Booking Assistant. 3. Improve Visibility of Pet Service Providers: Improved the visibility of small to medium-sized pet care service providers by giving them a dedicated platform to showcase their services and directly connect pet owners. Onboard and verify one (1) service provider within the first three (3) months of release. 4. Provide Business Insights for Service Providers: Enable pet care service providers to track monthly sales, customer feedback, and

	related insights by integrating a business analytics dashboard and AI-powered Data Analyst into the platform by August 2026.
SCOPE	The purpose of furlink is to innovate the way pet grooming services are booked and managed. The project meets the need for digital solutions in the local pet care market to reduce scheduling conflicts, enhance customer satisfaction, and improve service visibility for small to medium-sized pet grooming shops. Successfully deploy all planned release versions (v1.0 through v2.1) between April 2026 and August 2026, secure validation from academic advisers, and effectively facilitate bookings between pet owners and target service providers. High level risks for this project include ensuring user adoption among traditional service providers and pet owners who may be new to digital solutions and managing the technical implementation of AI features within the constraints of a student-led budget and timeline.
DELIVERABLES	The project deliverables shall include a web-based booking system that includes a calendar-based scheduling system, clear booking process, an AI booking assistant, business dashboards, and feedback.
SUCCESS CRITERIA	• AI Haircut Preview: The AI tool must be ready by August 2026, hitting at least 70% accuracy when showing what a pet's haircut will look like based on its current fur condition. • Zero Double-Bookings: Because of the AI Booking Assistant, no grooming shop should experience a single double-booking during the first 3 months after launch. • Service Provider Onboarding: Successfully verify, activate, and get at least 1 local grooming shop live on the app within 3 months of launching.

	• Active Storefronts: 100% of onboarded shops must have a working, public profile page so pet owners can find and contact them immediately. • Business Dashboard: Build a working sales and feedback tracking chart for service providers by August 2026. • AI Data Analyst: Deploy the AI Data Analyst by August 2026 so shop owners can automatically get summaries of their monthly performance. • Instant Payments: Ensure PayMongo payment links are generated instantly and without errors the moment a booking is confirmed. • Final App Sign-Off: Finish all user testing by August 31, 2026, with all critical bugs fixed and a user friendliness score of at least 3.5 out of 5 stars.
TARGET AUDIENCE	Dog/Cat Owners and Pet Service Providers The project includes dog or cat owners in that seeks new service options and are open to digital solutions. Another inclusion is small to medium-sized dog or cat grooming service providers that are locally based and not located in well-known areas.

Stakeholder Management Strategy

Introduction

The Stakeholder Management Strategy for furlink will be used to identify and classify project stakeholders; determine stakeholder power, interest, and influence; and analyze the management approach and communication methodology for project stakeholders. This will allow us to identify key influential stakeholders to solicit input for project planning and gain support as the project progresses. This will benefit the project by minimizing the likelihood of encountering competing objectives and maximizing the resources required to complete the project.

Early identification and communication with stakeholders are imperative to ensure the success of furlink by gaining support and input for the project. Some stakeholders may have interests which may be positively or negatively affected by the project.

By initiating early and frequent communication and stakeholder management, we can more effectively manage and balance this interest while accomplishing all project tasks.

Identify Stakeholders

The team will conduct a multi-layer discovery methodology to identify all parties influenced by or capable of influencing furlink. This process was designed to mitigate the risk of hidden stakeholders, which could otherwise result in scope creep or deployment delays. The identification process followed three primary phases:

- **Market Research:** The team will conduct an environment scan of the pet care grooming industry to identify the primary economic actors, including service providers or the grooming salons and the consumer base or the pet owners, as well as indirect influencers such as market competitors.
- **Brainstorming:** Internal sessions are held to map out the organizational and academic landscape. This helps identifying internal stakeholders—such as the Project Manager, Full-Stack Developers and Quality Assurance—as well as institutional stakeholders like the Project Advisers from Asia Pacific College.
- **Interviews:** To move beyond theoretical identification, the team conducts targeted interviews. This step serves as a verification mechanism to confirm that the identified users' needs aligned with the project's objectives and uncover secondary stakeholders, such as marketing personnel, who are essential for the platform's long-term viability.

Stakeholder Inclusion Criteria

The team utilizes the following criteria to determine if an individual or organization should be included as a stakeholder:

1. Is the person or their organization directly or indirectly affected by furlink?
2. Does the person or their organization hold a position from which they can influence project decisions?
3. Does the person have an impact on project resources such as technical tools, personnel, or academic funding?
4. Does the person possess specialized skills or technical capabilities required by the project?
5. Does the person potentially benefit from the project or are they able to resist the transition to a digital booking system?

Any individual who meets one or more of the above criteria is formally identified as a stakeholder.

Key Stakeholders

As a follow-on to the identification process, the team will identify a sub-set of stakeholders categorized as Key Stakeholders. These individuals or groups are determined by their significant influence over project outcomes, the extent to which they are affected by the platform, or their potential resistance to the transition from manual to digital booking system.

Because these key stakeholders require the most intensive communication and active management, the Project Manager is responsible for establishing a customized engagement plan. This plan will aim to obtain feedback on their desired level of participation and involvement in project milestones, their preferred frequency and method of communication, and any specific concerns, technical requirements, or conflicting interests that may arise during the Software Development Life Cycle.

Based on this gathered feedback, the Project Manager may involve these stakeholders in specific focus groups, beta testing reviews, or academic gate reviews to ensure the project remains aligned with both user needs and academic standards. Maintaining thorough communication with this group is essential to ensure that necessary resources remain available and that all project risks are addressed early.

Stakeholder Analysis

Once all the furlink stakeholders have been identified, the team categorizes and analyzes each individual and group. The purpose of this analysis is to determine the stakeholders' level of influence, plan the management approach, and to determine the appropriate levels of communication and participation each stakeholder will have on the project.

The team first categorizes stakeholders by their specific organization. Following this categorization, the team will utilize two primary tools to quantify and visualize stakeholder impact:

- **Power/Interest Matrix:** Used to illustrate the potential impact of each stakeholder by plotting them based on their level of authority or power and their level of concern or interest for project outcomes.

- Stakeholder Analysis Matrix: A tabular layout used to capture specific stakeholder concerns, their required level of involvement, and the designated management strategy based on their placement in the power/interest quadrants.

Power/Interest Matrix

Key	Organization	Name	Power (1-5)	Interest (1-5)
A	Management	Lalainne Andaya	5	5
B	Development Team	Atasha Frances Gayle Doria	4	5
C	Testing	Michelle Reina Pineda	4	5
D	Development Team	Feliz Angelica Salting	3	5
E	Faculty	Mr. Gonzalo Gumogda	4	4
F	Faculty	Mr. Gil Ballesca	3	4
G	Users	Service Providers	3	3
H	Users	Pet Owners	3	3
I	External	Competitors	1	2

Stakeholder Analysis Matrix

POWER	KEEP SATISFIED - Project Adviser - Project Co-Adviser	KEY PLAYERS - Project Manager - Lead Full-Stack Developer - Quality Assurance - Full-Stack Developer
	MINIMAL EFFORT Competitors	KEEP INFORMED - Service Provider - Pet Owner
INTEREST		

Based on the power and interest analysis and chart above, stakeholders E and F, in the upper left quadrant, must be kept satisfied by providing a high-quality documentation and regular milestone updates to ensure all academic requirements are met. On the top right quadrant are stakeholders A, B, C, and D or the key players who are involved in all levels of project planning, technical development, and marketing. With that, the roles requires the team to hold daily stand-up meetings to ensure alignments. Stakeholder I on the lower left quadrant needs to be monitored regularly for market trends, requiring minimal direct management effort. Lastly, stakeholder G and H on the lower right quadrant must be kept informed through progress updates and beta-testing opportunities to ensure the final product meets market needs.

The stakeholder analysis matrix will be used to capture stakeholder concerns, level of involvement, and management strategy based on the stakeholder analysis and power/interest matrix above. The stakeholder analysis matrix will be reviewed and updated throughout the project's duration to capture any new concerns or stakeholder management strategy efforts.

Stakeholder	Concerns	Quadrant	Management Strategy
Project Manager	Balancing administrative milestones, coordinating team schedules, managing project resource limits, and overseeing marketing outreach.	Key Player	Maintain absolute involvement in all levels of system planning and operational change management. Hold structured internal sync-up sessions to keep timelines unified.
Lead Full-Stack Developer	Technical architecture scalability, database normalization, smooth third-party API integrations, and maximizing system security.	Key Player	Actively participate in all development status sprints and milestone design reviews. Collaborate directly with the Project Manager to align codebase execution with timeline goals.
Quality Assurance	Minimizing runtime execution bugs, validating frontend field input safety, and verifying that all technical aspects match core user stories.	Key Player	Engage deeply in feature deployment reviews. Provide immediate, continuous automated error feedback loops to the development branch to resolve critical bugs early.

Full-Stack Developer	Resolving system architecture technical debt, implementing client-side page transitions, and optimizing background edge worker speeds.	Key Player	Maintain daily development contributions and active integration in all sprint planning sessions. Co-author comprehensive code.
Project Adviser	Proper adherence to the Software Development Life Cycle and meeting all Asia Pacific College academic requirements.	Keep Satisfied	Communicate project specifications and academic deliverables as required to ensure institutional satisfaction.
Project Co-Adviser	Proper adherence to the Software Development Life Cycle and meeting all Asia Pacific College academic requirements.	Keep Satisfied	Communicate project specifications and academic deliverables as required to ensure institutional satisfaction.
Service Providers	Smooth transitions from manual scheduling methods to digital interfaces, receiving timely service payouts, and the accuracy of AI data analytics reports.	Keep Informed	Provide status updates and obtain feedback by providing targeted hands-on onboarding sessions and dashboard walkthroughs.
Pet Owner	Complete data privacy of personal/pet medical records, marketplace profile visibility, and an intuitive, fast booking and checkout workflow.	Keep Informed	Maintain continuous communication regarding application safety parameters and performance features. Solicit end-user feedback through dedicated beta-testing opportunities.
Competitors	Loss of local user acquisition momentum to furlink's highly tailored,	Minimal Effort	Execute passive tracking of market landscape updates and feature trend shifts through periodic secondary

	localized, and niche-specific AI-driven features.		research. No direct engagement required.
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Stakeholder Analysis

Name	Department / Company	Position	Advisers	Objectives, Requirements, Interests	Influence	Project Contribution	Resistance
Lalaine Andaya	Management	Project Manager	Project Advisers, Lead Full-Stack Developer, Quality Assurance, Users	Deliver project on schedule, monitor budgets, ensure optimal resource allocation, and guide market outreach.	High	Project planning, milestone coordination, risk mitigation tracking, and market research leadership.	Resists expanding application scope (scope creep) to prevent budget overruns.
Atasha Frances Gayle Doria	Development Team	Lead Full-Stack Developer	Project Manager, Project Advisers, Full-Stack Developer, Quality Assurance	Establish scalable cloud-native serverless architecture, secure Supabase integrations, and ensure low-latency API performance bounds.	High	Develops and deploy/leads platform architecture, coding, and ensure system security	Resists implementing non-essential features early to prioritize resolving core architectural technical debt.
Michelle Reina Pineda	Testing	Quality Assurance	Project Manager, Project Advisers,	Ensure the system is bug-free, meets all requirements	High	Leads system testing and bug reporting.	Resists premature feature deployment

			Lead Full-Stack Developer	and build brand awareness.			handovers until strict boundary testing and validation criteria are fully satisfied.
Feliz Angelic Salting	Development Team	Full-Stack Developer	Lead Full-Stack Developer	Develop the system's features and meet all technical goals.	High	Develops system modules.	Resists implementing non-essential feature.
Mr. Gonzalo Gumogda	Faculty	Project Adviser	Project Manager, Full-Stack Developer, Quality Assurance	Ensure strict compliance with institutional Systems Development Life Cycle methodologies and prompt fulfillment of all academic grading deliverables.	High	Academic governance oversight, formal milestone gate evaluations, and high-level project strategic guidance.	Resists project progression if there is any deviation from standard institutional criteria or incomplete documentation templates.
Mr. Gil Ballesca	Faculty	Project Co-Adviser	Project Manager, Full-Stack Developer, Quality Assurance	Ensure strict compliance with institutional Systems Development	Medium	Academic governance oversight and high-level project strategic guidance.	Resists project progression if there is any deviation from standard institutional

				Life Cycle methodologies and prompt fulfillment of all academic grading deliverables.			criteria or incomplete documentation templates.
Service Providers	Users	Client	Project Manager	Expects increased sales from successful bookings, better market exposure, and comprehensible business reports.	Medium	Provides service listings and essential feedback for B2B features.	Hesitates and resists transitioning operational workflows away from familiar process due to digital navigation friction.
Pet Owners	Users	Client	Project Manager	Expects a hassle-free booking experience with trusted service providers.	Medium	Participate in multi-stage beta testing windows, submit structured user feedback, and provide verified pet health/vaccination records.	Hesitates and resists transitioning operational workflows away from familiar process due to digital navigation friction.

Competitors	External Market Platforms			Continuously compete in the growing market, providing opportunities for innovation.	Low	Act as a baseline control group for local industry feature gap assessments and secondary competitive market research.	Actively counters furlink's market entry by leveraging existing user loyalty and matching core automated features.
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Statement of Work

Introduction

furlink is a student-led startup project developed by the team LogiTech from Asia Pacific College. It aims to modernize the way pet grooming services are booked and managed in the Philippines.

Pet ownership in the Philippines has rapidly grown with approximately 16.4 million pet dogs recorded in 2020. Despite the rapidly expanding pet care industry with an estimated \$1.2 billion USD market as of 2025, service providers in Metro Manila continue to rely on informal booking methods such as Facebook Messenger, Instagram, or through phone calls. This results in scheduling conflicts, poor visibility for small-scale pet service providers, and unmet expectations for pet owners.

The platform directly addresses these challenges by providing a centralized, web-based booking system that specializes in connecting dog or cat owners with pet grooming service providers through a streamlined digital platform.

This Statement of Work (SOW) defines the scope, deliverables, schedule, acceptance criteria, and responsibilities for development, testing, and the deployment of furlink.

Scope of Work

The scope of this project includes all planning, execution, testing, and deployment of the web-based booking platform. The platform will serve two primary user types: dog or cat owners and pet grooming service providers. The platform will include the following:

1. User Management
 - Development of registration and login systems for pet owners and pet grooming service providers.
 - Implementation of a provider onboarding and verification workflow to authenticate new partners.
2. Booking Functionality
 - Implementation of a centralized booking system and service listing tools.
 - Integration of an AI Booking Assistant designed to manage calendar availability in real-time and actively prevent double-booking incidents.
3. AI-Powered Grooming Preview

- Development and integration of an AI-powered haircut preview tool.
- 4. Business Intelligence
 - Integration of business analytics dashboards to help service providers track revenue, growth, and customer feedback.
 - Integration of an AI-powered Data Analyst to interpret dashboard metrics and generate actionable business insights for the providers.

Not included in the scope of work of this project are the following:

1. Other Pet Service
 - furlink will not cater to booking or management of veterinary care, pet supplies, boarding, and pet transportation services
2. Payment Data Storage
 - In-depth financial reporting or internal accounting tools for businesses.
 - The system will not collect or store raw credit card details; all financial handovers are processed exclusively via external secure payment links.
3. Financial Auditing
 - The platform will not provide in-depth financial reporting, tax preparation, or internal accounting tools for the service providers' broader business operations.
4. Foundational AI Model Training
 - The project will utilize and fine-tune existing AI APIs for the haircut preview and data analysis features, rather than building foundational artificial intelligence models from scratch.

Period of Performance

The performance period for the furlink platform is six (6) months (180 days) beginning on March 7, 2026, through October 1, 2026. All work must be scheduled to be completed within this timeframe. Any modifications or extensions will be requested through the Project Manager and the Project Adviser for review and discussion.

Place of Performance

The team for the furlink project will perform the majority of the work at their own respective workstations. This arrangement supports the team's use of cloud-based development and collaboration tools, including Vercel for hosting, GitHub for version

control, and OpenProject for task management. However, the team will be required to meet on premises once per week for a weekly status meeting within the team and with the Project Adviser. All on-premises meetings will be held at school campus, but in times of unavailability of consultations rooms, co-working spaces will be provided by the team. Once the project has been completed and is in the process of onboarding pet grooming service providers, all meetings with the client will be held online or on premises as discussed between them.

Work Requirements

Project Kickoff

- Present and finalize Statement of Work with Stakeholders
- Establish sprint schedule using Agile/Scrum methodology
- Finalize team roles and responsibilities
- Set up development environments and project management tools

Design Phase

- Gather and document functional and non-functional requirements from pet owners and service providers
- Produce wireframes and UI mockups via Figma for all platform screens
- Define the Entity Relationship Diagram (ERD), Data Flow Diagrams, and Use Case Diagrams
- Present design artifacts to the project adviser and stakeholders for review and approval
- Finalize the Product Backlog, Product Roadmap, and Release Plan

Development Phase

- Complete all development for approved design
- Accomplish a detailed test plan
- Conduct testing in different devices and environments
- Present status during daily scrums

Testing Phase

- 100% of unit test cases must pass
- Code completion at minimum 80%
- All key modules verified for end-to-end data flow

- No critical bugs in staging environment
- User acceptance testing conducted by at least one on-board service provider
- Resolve any coding issues identified in testing
- Provide written test case reports to stakeholders

Deployment Phase

- Deploy furlink v2.0 to Vercel cloud hosting
- Onboard and verify at least 3 pet grooming service providers
- Monitor platform performance and resolve post-deployment issues
- Provide weekly status updates to project adviser

Project Close

- Provide complete project documentation
- Submit final project closure report
- Complete project requirements checklist

Schedule/Milestones

The list below consists of initial milestones identified for the furlink project:

Milestones	Date
Capstone 2 Kickoff	March 7, 2026
Final Testing	July 6-7, 2026
Release to Service Provider (Training)	July 8, 2026
furlink v1.0 Deployment	July 11, 2026
Enhancement of Core Booking Modules	July 13-17, 2026
Development of AI Booking Assistant	July 20-30, 2026
System and Integration Testing	July 31, 2026 – August 7, 2026
furlink v2.0 Release	August 10, 2026
User Acceptance Testing & Fixes	August 17, 2026 – August 21, 2026
Solution Release furlink v2.1	August 31, 2026
Project Completion	October 1, 2026

Acceptance Criteria

The project will be accepted as complete when it successfully passes the following testing milestones overseen by the QA Lead:

- Unit Testing: Entry: Individual functions/modules developed and peer-reviewed.

- Exit: 100% of unit test cases pass, no critical/major bugs, code is at least 80% complete.
- Integration Testing: Entry: Key modules (registration, booking flow, AI generator, calendar sync, dashboard) pass unit testing and are connected.
- User Acceptance Testing (UAT): Entry: System is complete, stable, and ready for stakeholder testing.

Other Requirements

Security and Data Privacy

The team members must maintain secure access to the project's GitHub repository, Supabase backend, and hosting environment. The platform shall comply with the Data Privacy Act of 2012 (Republic Act No. 10173) of the Philippines. All user data will be collected, stored, and processed only with informed consent. Appropriate technical and organizational security measures will be implemented.

Resource Constraints

Because this project is student-led with limited financial resources, development will prioritize free-tier and open-source tools to manage the estimated budget.

Business Model Canvas

KEY PARTNERS	KEY ACTIVITIES	VALUE PROPOSITIONS	CUSTOMER RELATIONSHIPS	CUSTOMER SEGMENTS
Pilot Service Providers or local grooming shops that will be important	Platform development : Full-stack development API integration:	Furlink provides a centralized digital booking platform for the pet care market For pet owners, furlink provides a convenient way for booking pet grooming services with the help of the AI chatbot to assist. Another help are the	The AI booking assistant will handle inquiries 24/7, including scheduling and FAQ resolutions. The application admin will be responsible for verifying the	Furlink operates on a B2B2C model, bridging local businesses with end

for the initial testing. On the other hand, academic help and financial sponsors are vital to the project. This includes APC-SoCIT advisers and the project group's financial contribution to furlink.	Embedding Gemini 2.5 flash and PayMongo for the in-app payment transactions. Service Provider Onboarding: Verifying and setting up digital profiles for local grooming shops. Quality Assurance: Executing unit testing and User Acceptance Testing (UAT). Platform Maintenance: Monitoring system uptime and managing	transparent pricing and verified shop reviews Furthermore, service providers are capable of eliminating double-bookings through the real-time calendar sync and there's protection against no-shows through the PayMongo downpayments. Additionally, there's a reduced need for their respective admin for assisting the pet owner. Lastly, service providers have access to the business dashboard to track sales and feedback.	service provider applications and managing any platform problems.	consumers . Segment 1 (B2B): Pet grooming shops that currently rely on manual booking processes and aim for operational efficiency, digital visibility, and protection from missed appointments will be onboarded to the furlink platform Segment 2 (B2C): Dog and cat owners in Makati City seeking a
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	database security (RLS).			reliable, structured, and visually
	KEY RESOURCES Project Manager Lead Full stack developer Full stack developer QA lead Furlink codebase GitHub repository System architecture design Foundational project budget of ₱675,332.08		CHANNELS Direct outreach to local grooming shops. Social Media Posting. The furlink web application which will be accessible for both mobile and desktop devices.	communicative way to book pet care services will be encouraged to use the furlink platform.
COST STRUCTURE		REVENUE STREAMS		
furlink operates on a value-driven financial model, divided between initial system development and		furlink prioritizes user acquisition and system validation over immediate profit for the first few months of deployment. The future scalability model includes:		

ongoing operational scaling • Development: Authorized budget of ₱420,374.26 required to design and deploy the system. • Operational Expenditures (Year 1): Annual costs of ₱61,821.79 sustain the platform post-launch.	• Primary Stream: A percentage-based commission or fixed convenience fee per successful booking processed through PayMongo to be implemented in the future. • Secondary Stream: Promoted/Sponsored listings within the Furlink directory to boost SP visibility.
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Feasibility Study

Executive Summary

furlink is a startup designed to digitize the fragmented Philippine pet grooming industry by replacing inefficient, informal booking methods like Facebook Messenger with a structured digital ecosystem. The platform addresses critical market gaps by providing a centralized web-based interface that eliminates scheduling conflicts and provides much-needed business visibility for local service providers.

The platform offers pet owners transparent pricing and a seamless booking experience enhanced by an AI-powered Booking Assistant for automated scheduling. Service providers gain a professional online presence and a Business Intelligence Dashboard that delivers actionable insights into sales performance and customer feedback to drive business growth.

Built on a scalable Backend-as-a-Service infrastructure using React.js, Supabase, and Vercel, the platform is managed by a four-person core team, ensuring minimal technical overhead. Following a targeted pilot in Makati City, furlink will transition to a commercial model funded by transaction fees and promotional campaigns, positioning it to capture a significant share of the growing pet care market.

Description of Products and Services

furlink is a centralized digital booking platform designed to streamline the pet grooming experience for both pet owners and grooming service providers. The platform addresses a widely recognized gap in the local pet care industry, specifically the absence of a dedicated, structured booking system, by replacing informal scheduling methods—such as Facebook Messenger and phone calls—with an integrated digital ecosystem.

For pet owners, furlink provides a seamless experience for discovering nearby grooming service providers, viewing service offerings and pricing, scheduling appointments, and managing booking history, all within a single platform. The platform also integrates an AI-powered Booking Assistant that guides users through the booking process conversationally, automatically extracting pet and appointment details from natural language input to reduce friction and accelerate the scheduling experience.

For service providers, furlink offers a professional online presence with a dedicated listing page, a structured application and onboarding process, and real-time booking management tools. To drive business growth, the platform features a comprehensive Business Intelligence Dashboard coupled with an AI-powered Data Analyst. While the dashboard provides visual summaries of sales performance, booking trends, and customer feedback, the integrated AI Data Analyst interprets these metrics automatically. Acting as an on-demand business consultant, this AI feature processes historical transaction and scheduling data to generate proactive recommendations. This provides non-technical grooming business owners with actionable, data-driven insights without requiring manual data analysis.

furlink designed as a local-first platform, initially focusing on pet grooming services within Makati City, with a structured plan to expand its coverage to additional cities and service categories as platform adoption grows. No changes to the existing service offerings of partner grooming businesses are required as part of this initiative, as furlink functions as the booking, management, and intelligent analytics layer on top of their existing operations.

Technology Considerations

furlink is a web-based platform requiring a modern, scalable infrastructure to support its core business functions, including booking management, AI-assisted user interactions, and secure payment processing. To minimize initial development costs and technical risk, the team has strategically adopted a Backend-as-a-Service (BaaS) strategy,

leveraging established cloud-based platforms rather than constructing physical infrastructure from scratch. The platform is built using React.js to ensure a responsive user experience across devices, with development and automated deployment pipelines managed through Vercel to maintain high availability without the need for manual server maintenance. Backend operations, including user authentication and database management, are centralized in Supabase using PostgreSQL, which provides enterprise-grade security through Row Level Security and serverless Edge Functions for executing background logic.

The AI Booking Assistant is powered by the Google Gemini API and orchestrated via the Vercel AI SDK, utilizing the core model's native function-calling capabilities to seamlessly parse natural language conversational inputs into structured database entries. To address regulatory and security risks regarding financial transactions, PayMongo is integrated as the primary payment gateway, offering a PCI-compliant environment for local Philippine payment methods without requiring the platform to handle sensitive card data directly. Furthermore, transactional notifications are automated through Gmail SMTP and triggered by specific system events to ensure consistent real-time engagement with users.

Beyond conversational support, the platform features an AI Pet Haircut Generator that relies on dedicated Hugging Face serverless inference endpoints to execute state-of-the-art open-source diffusion models via dedicated cloud GPU compute. To ensure absolute uptime and shield the user experience from potential infrastructure bottlenecks, the system architecture utilizes Pollinations.ai as an immediate, asynchronous failover routing engine. Additionally, an AI-Powered Data Analyst is implemented using Gemini 2.5 Flash, leveraging its massive 1-million token context window to securely ingest and process entire historical datasets, anonymized booking trends, and platform metrics in a single stateless execution payload. This configuration provides service providers with enterprise-grade business intelligence and predictive demand forecasting without violating user data privacy or incurring prohibitive computational overhead.

The final analysis indicates that this technology stack consists of proven, industry-standard tools, and because the student team possesses the internal expertise to maintain these systems, there is no requirement for external development contracts, ensuring a sustainable cost-to-growth ratio for the startup as it transitions into progressive commercial scaling.

Product/Service Marketplace

The pet care industry in the Philippines represents a rapidly growing market driven by urbanization and a cultural shift toward treating pets as a members of the family. A significant portion of Filipino pet owners now allocate a meaningful share of their household budget to pet care services, with grooming being among the most frequently and consistently demanded services. This sustained demand presents a clear and viable market opportunity for a dedicated digital booking platform.

Despite this growth, the booking process for pet grooming services in the Metro Manila area remains largely informal. A recent survey found that the majority of small-scale pet service providers rely almost exclusively on Facebook Messenger and direct phone calls to manage appointments, leading to scheduling inefficiencies, double-booking incidents, and a general lack of business visibility. Pet owners, on the other hand, face difficulty comparing service providers, viewing accurate pricing upfront, and managing their booking history in a centralized manner. furlink directly addresses these pain points.

Target Market

furlink's primary target market consists of two groups. The first is pet owners in Makati City, specifically those who own dogs and cats and regularly seek professional grooming services. This group values convenience, pricing transparency, and trust in the service providers they engage. The second is small to medium-sized pet grooming businesses operating in Makati City, particularly those that currently rely on traditional booking channels and would benefit from a structured digital presence, automated scheduling, and business performance insights.

Competitive Landscape

furlink operates in a space where the primary competition comes not from direct booking platform rivals, but from the informal channels that currently dominate the market. The most significant competitors are Meta platforms, particularly Facebook and Facebook Messenger, which pet owners and service providers habitually use to discover and coordinate grooming appointments. The conversational and familiar nature of these channels presents a behavioral inertia that furlink must overcome through ease of use and clear added value.

Among existing digital platforms, PetBacker is the closest comparable service, offering a marketplace for connecting pet owners with service providers across Southeast Asia. Nevertheless, PetBacker has documented limitations including a lack of upfront pricing transparency and requirement for users to submit pet information before they can

explore the platform, which creates friction for new users. Rover, while a well-established international platform, is limited to the United States and Canada, is therefore not a direct competitor in the Philippine market. No existing platform currently offers a locally tailored, AI-assisted booking experience with an integrated business dashboard for providers.

Distribution and Delivery

furlink is a fully digital platform accessible via web browser on any internet-connected device. Service discovery, booking, payment, and communication all occur within the platform. The actual grooming services are delivered physically at the service provider's location. furlink serves as the coordination and transaction layer, not a service delivery mechanism. This asset-light model requires no warehousing, logistic infrastructure, or physical facilities beyond the platform itself.

Why Pet Owners Would Choose furlink

Pet owners are expected to choose furlink over informal booking channels because the platform's pricing transparency, ease of scheduling, centralized booking history, verified provider listings, and the convenience of the AI Booking Assistant, which simplifies the booking process significantly compared to back-and-forth messaging platforms.

Marketing Strategy

furlink's go-to-market strategy is designed for a startup environment with a lean budget, focusing on targeted digital marketing and a controlled platform launch with a single verified service provider partner.

Launch Strategy: Single Provider Pilot

For the initial launch phase, furlink will partner exclusively with Groom Room by Steve, a grooming service provider that the team has coordinated with directly for platform testing and validation. This controlled approach allows the team to validate the platform's core booking workflow, gather user feedback, and demonstrate furlink's value proposition in a live environment before scaling to additional providers.

Social Media Marketing Campaign

furlink will leverage social media platforms, primarily Facebook and Instagram, to drive awareness and encourage pet owners to book their first appointment through the platform. The campaign will center on furlink's seamless user experience, the AI Booking Assistant feature, and the ability to discover and book grooming services in one platform.

As a highlight of the launch, furlink will introduce a promotional initiative designed to honor and reward pet owners who support the platform with early bookings. To foster a high-energy community and incentivize first-time usage, winners will be selected through a lucky draw mechanism to receive exclusive cash rewards.

Differentiation

furlink differentiates itself from existing alternatives in three keyways:

1. It is designed specifically for the local pet care market, making it more relevant and culturally aligned than international platforms.
2. It incorporates an AI Booking Assistant, which represents meaningful innovation over competitors who offer no AI-assisted functionality.
3. It provides service providers with a Business Intelligence Dashboard that gives grooming businesses visibility into their performance, a feature not available through traditional booking channels or competing platforms.

Being a student-developed startup also gives furlink an authentic community narrative that resonated well in local social media marketing.

Future Marketing Considerations

As the platform grows beyond the pilot phase, furlink plans to expand its provider listings and introduce paid promotional programs for service providers. These programs would feature participating grooming businesses in platform-wide campaigns, giving providers greater visibility in exchange for a sponsorship or campaign fee. This creates a secondary revenue stream while simultaneously giving the platform additional marketing content and service provider engagement.

Organization and Staffing

The implementation, deployment, and ongoing governance of the furlink platform are managed by a specialized, cross-functional four-person core startup team. This internal team possesses the combined expertise in full-stack software engineering, quality assurance, and agile project management necessary to construct and maintain the cloud-based infrastructure without requiring external technical contracts or high-cost development consultancies. The organizational structure operates on a flat, collaborative framework engineered to maximize velocity during parallel development sprints. A dedicated internal labor cost is systematically computed across all project phases to account for engineering sprints, quality benchmarks, and operational governance. Consequently, no external hierarchical additions are required during the implementation period, ensuring a sustainable and agile cost-to-growth ratio as the platform transitions from an academic project into a commercial startup.

The core startup team and its governance body are comprised of the following designations:

- **Lalaine Andaya (Project Manager and Full-Stack Developer):** Oversees overall project coordination, milestone tracking, and sprint scheduling, while simultaneously contributing directly to the platform's codebase by developing assigned functional modules.
- **Atasha Frances Gayle Doria (Lead Full-Stack Developer):** Orchestrates the overarching system architecture, root database schemas, security configurations, and primary payment gateway integrations, while supervising the functional implementation of the conversational AI components.
- **Michelle Reina Pineda (Quality Assurance):** Responsible for the rigorous execution of regression, usability, and cross-device compatibility testing pipelines, validating that all code deployments meet the platform's stringent quality benchmarks before entering production.
- **Feliz Angelica Salting (Full-Stack Developer):** Responsible for end-to-end full-stack engineering of assigned platform modules, writing responsive React.js user interfaces, and constructing corresponding backend logic and database workflows.
- **Change Control Board (CCB):** Functions as a collective governing body composed of all four core team members to manage configuration repository branches,

evaluate scope alterations, and filter proposed system modifications against technical feasibility and strict deadlines.

Schedule

furlink development and release schedules spans from the Capstone 2 kickoff through project completion, covering the full lifecycle from platform final testing to solution release. The following milestones represent the high-level implementation timeline for the project:

Milestones	Date
Capstone 2 Kickoff	March 7, 2026
Final Testing	July 6-7, 2026
Release to Service Provider (Training)	July 8, 2026
furlink v1.0 Deployment	July 11, 2026
Enhancement of Core Booking Modules	July 13-17, 2026
Development of AI Booking Assistant	July 20-30, 2026
System and Integration Testing	July 31, 2026 – August 7, 2026
furlink v2.0 Release	August 10, 2026
User Acceptance Testing & Fixes	August 17, 2026 – August 21, 2026
Solution Release furlink v2.1	August 31, 2026
Project Completion	October 1, 2026

Financial Projections

The financial projections for furlink are structured around a phased deployment strategy designed to capture initial market share and maximize user acquisition before transitioning into a self-sustaining commercial model. These projections comprehensively account for long-term cloud infrastructure maintenance, targeted marketing expenses, and the calculated transition from an academic pilot into a live, dual-sided marketplace.

Revenue Model

To establish market trust and eliminate initial friction for both pet owners and service providers, furlink operates under a multi-stage monetization timeline:

- **The Pre-Revenue Phase:** During this initial stage, the platform's core booking infrastructure is fully operational, but transactions are completely non-monetized

or 0% platform commission. This strategic choice allows the team to validate system workflows, eliminate backend operational risks, and generate the critical network effects necessary to build a highly active community of pet owners and service providers without financial barriers to entry.

- **The Commercialization Phase:** Upon achieving operational stability and a baseline user network post-pilot, furlink will activate its commercial framework, introducing two primary, synergistic revenue streams:
 - **Transaction-Based Marketplace Commissions:** furlink will transition into a monetized booking layer, generating its primary revenue by collecting a fixed percentage-based commission from each successfully completed grooming appointment facilitated through the platform's escrow logic.
 - **Featured Provider Campaigns:** To supplement transaction revenues, furlink will launch an opt-in, voluntary promotional program for service providers. Participating service providers can purchase featured visibility slots to be prominently displayed at the top of the user interface and integrated into furlink's targeted social media marketing channels, providing small businesses with a high-conversion mechanism to accelerate their client growth.

Cost Figures

Measure	Year 1	Year 2	Year 3	Year 4	Year 5
Projected Revenue	₱56,590.75	₱150,163.15	₱247,426.13	₱361,457.36	₱488,688.60
Hosting Infrastructure	₱38,449.87	₱46,245.12	₱54,897.84	₱65,561.25	₱78,188.48
System Maintenance	₱23,371.92	₱24,540.52	₱25,767.54	₱27,055.92	₱28,408.71
TOTAL OPERATIONAL COSTS	₱61,821.79	₱70,785.64	₱80,665.38	₱92,617.17	₱106,597.20
Net Cash Flow	-₱5,231.04	₱79,377.51	₱166,760.75	₱268,840.19	₱382,091.40

Findings and Recommendations

Based on the comprehensive research, milestone scheduling, and operational analysis presented in this feasibility study, it is recommended that the organization approves the formal initiation, deployment, and market launch of the furlink platform. The findings indicate that the initiative is highly viable, addresses a critical data gap in the local pet care ecosystem, and possesses a high probability of technical and commercial success within the target region. Key findings are summarized below:

Technology

- **Infrastructure Efficiency:** The project's strategic adoption of a Backend-as-a-Service (BaaS) model—leveraging high-availability cloud routing through Vercel and centralized PostgreSQL database instances on Supabase—effectively eliminates technical risks and removes the financial burden of constructing physical server infrastructure from scratch.
- **Innovation:** The integration of the Google Gemini-powered AI Booking Assistant and AI-powered Data Analyst provides a unique competitive edge by automating pet detail extraction and service discovery.
- **Resilient Architecture:** Incorporating a server-side multi-model graphic rendering pipeline for the AI Pet Haircut Generator—utilizing a high-performance model on a Hugging Face Inference Endpoint with automated failover routing to Pollinations.ai—guarantees absolute platform uptime and consistent user experiences.
- **Internal Capability:** The internal startup team possesses the necessary technical expertise to implement and maintain the stack, removing the requirement for expensive external development contracts.

Marketing

- **Market Opportunity:** There is a clear, culturally driven demand for professional grooming services, which is currently underserved by highly inefficient and informal coordination methods like Facebook Messenger and direct phone calls.
- **Differentiation:** furlink successfully differentiates itself from existing international platforms through complete upfront pricing transparency, a localized community narrative, and a dedicated Chart.js-powered Business Intelligence Dashboard that provides small B2B salons with unprecedented operational visibility.
- **Controlled Launch Scalability:** The go-to-market strategy uses an asset-light, single-provider pilot phase with Groom Room by Steve, enabling low-risk workflow validation, early user feedback loops, and targeted local social media campaigns before scaling to additional cities.

Organizational

- **Self-Sufficiency:** The four-person core startup team is structured effectively to handle development, operations, and marketing within a collaborative framework.
- **Minimal Overhead:** No organizational restructuring or capital investment in new physical facilities is required, maintaining an agile startup posture.

Financial

- **Asset-Light Model:** By leveraging internal engineering labor and utilizing cloud technology infrastructure tiers during the academic phase, the project successfully avoids immediate capital constraints and maintains an optimized cost-to-growth ratio.
- **Future Viability:** The multi-stage monetization timeline—safely moving from a 0% commission, zero-revenue academic pilot to a commercialization phase driven by transaction-based marketplace commissions and featured provider promotional campaigns—establishes a clear, sustainable path toward long-term profitability.

furlink project is technically sound, operationally realistic, and strategically aligned with dominant pet consumer trends in the Philippines. By streamlining the coordination and transaction layers through automated serverless triggers, innovative AI modules, and dynamic data insights, the platform is uniquely positioned to capture meaningful market share and deliver high structural value to the local pet care ecosystem.

Planning

Communications Management Plan

Introduction

This Communications Management Plan sets the communications framework for this project. It will serve as a guide for communications throughout the life of the project and will be updated as communication needs change. This plan identifies and defines the roles of people involved in this project. It also includes a communications matrix which maps the communication requirements of this project. An in-depth guide for conducting meetings details both the communications rules and how the meetings will be conducted, ensuring successful meetings. A project team directory is included to provide contact information for all stakeholders directly involved in the project.

Communications Management Approach

The Project Manager will take a proactive role in ensuring effective communications on this project. The communications requirements are documented in the Communications Matrix presented in this document. The Communications Matrix will be used as the guide for what information to communicate, who is to do communication, when to communicate, and to whom to communicate.

As with most project plans, updates or changes may be required as the project progresses or changes are approved. Changes or updates may be required due to changes in personnel, scope, budget, or other reasons. Additionally, updates may be required as the project matures, and additional requirements are needed. The project manager is responsible for managing all proposed and approved changes to the communications management plan. Once the change is approved, the project manager will update the plan and supporting documentation and will distribute the updates to the project team and all stakeholders. This methodology is consistent with the project's Change Management Plan and ensures that all project stakeholders remain aware and informed of any changes to communications management.

Communications Management Constraints

All project communication activities will occur within the project's approved budget, schedule, and resource allocations. The project manager is responsible for ensuring that communication activities are performed by the project team without external resources, which will result in exceeding the authorized budget. Communication activities will occur in accordance with the frequencies detailed in the Communication Matrix in order to ensure the project adheres to schedule constraints. Any deviation of these timelines may result in excessive costs or schedule delays and must be approved by the project sponsor.

Everyone part of the project, must first obtain the approval of all the other members to authorize the distribution of confidential information. The project manager is responsible for ensuring that approval is requested and obtained prior to the distribution of any confidential information and materials regarding this project.

Stakeholder Communications Requirements

As part of identifying all project stakeholders, the project manager will communicate with each stakeholder in order to determine their preferred frequency and method of communication. Standard project communications will occur in accordance with the Communication Matrix; however, depending on the identified stakeholder communication requirements, individual communication is acceptable and within the constraints outlined for this project.

In addition to identifying communication preferences, stakeholder communication requirements must identify the project's communication channels and ensure that stakeholders have access to these channels. If project information is communicated via secure means or through internal resources, all stakeholders, except for Service Providers, must have the necessary access to receive project communications.

Once all stakeholders have been identified and communication requirements are established, the project team will maintain this information in the project's Stakeholder Register and use this, along with the project communication matrix as the basis for all communications.

Roles

Project Sponsor

The project sponsors for furlink consist of the project team members and their immediate family members who provide the foundational financial backing and logistical support for the project. As the primary funders, they are ultimately invested in its success. Communications to the sponsors should be presented in a summary format detailing project milestones, budget health, and major achievements, unless more detailed financial or scheduling updates are requested.

Project Adviser

In the context of this academic project, the Project Adviser fulfills the role of a program manager overseeing the project within the School of Computing and Information Technologies. The Project Adviser is responsible for ensuring that the project meets academic standards, course requirements, and timelines. They require more detailed, frequent communications than Project Sponsors, including progress reports, technical blockers, and updates.

Key Stakeholders

While many individuals are impacted by furlink, the Key Stakeholders for this specific communication plan include the peer reviewers, and early-adopter grooming service providers. These are the stakeholders who require targeted communication regarding beta testing schedules and deployment impacts, but are not involved in the day-to-day execution.

Change Control Board

For the furlink capstone initiative, the Change Control Board (CCB) functions as the definitive approval authority for all proposed changes to the project's scope, timeline, budget, or multi-cloud infrastructure. Chaired by the Project Adviser and Co-Adviser, and including the Project Manager and Lead Developer, this collaborative group rigorously evaluates technical specifications and requested modifications against strict academic deadlines. The CCB ensures that any authorized changes are absolutely essential to delivering the core MVP and do not introduce scope creep that could jeopardize the final solution release.

Project Manager

Responsible for the overall success of the Project. The PM must authorize and approve all project expenditures. The PM is also responsible for approving that work activities meet established acceptability criteria and fall within acceptable variances. The PM will be responsible for reporting project status in accordance with the communications management plan. The PM will evaluate the performance of all project team members. The PM must possess the following skills: leadership/management, budgeting, scheduling, and effective communication.

Lead Full-Stack Developer

Accountability for the end-to-end technical architecture, code security, and final deployment of the furlink web application. The LD holds the authority to make critical decisions regarding the core tech stack, external library integrations, database schema modifications, and deployment workflows. Primary responsibilities include designing the system architecture, setting up the Supabase database infrastructure, configuring third-party APIs (Gemini AI and PayMongo), and overseeing the repository's main code branch. This role requires advanced technical competency in Vite, Node.js, and PostgreSQL, along with strong leadership skills to guide code freezes, conduct code reviews, and cross-train team members on database conflict resolution.

Full-Stack Developer

Responsible for the core feature development, implementation, and maintenance of both the user interface and backend application logic. The FD has the authority to implement design patterns, write specific API queries, and construct reusable components within the established architectural guidelines. Primary responsibilities include translating user stories into responsive UI components using Vite/CSS, managing Supabase backend data queries, handling CRUD operations for provider listings, and building real-time dashboard analytics. This role requires technical proficiency in JavaScript, CSS, and

database manipulation, with a focus on ensuring the platform functions seamlessly across various mobile viewports and browser environments.

Quality Assurance Tester

The QA Tester focuses on ensuring the reliability and user-friendliness of the platform by developing comprehensive test plans that cover both manual functional testing and automated regression suites. This role holds the authority to flag critical bugs that could delay a release, ensuring that the software is thoroughly validated before it reaches the end user. Responsibilities include documenting all identified issues, verifying technical fixes, and facilitating User Acceptance Testing (UAT) sessions with actual pet service providers to measure the platform's ease of use. Success in this role requires strong analytical skills and a deep understanding of the Software Testing Life Cycle (STLC) to prevent data errors or UI failures.

Project Team Directory

Role	Name	Title	Organization/ Department	Email
Project Sponsor Representative	Manuel R. Doria	Sponsor Representative	LogiTeh	manueldoria14@yahoo.com
Project Adviser	G. Gumogda	Project Adviser	APC-SOCIT	gonzalog@apc.edu.ph
Project Manager	Lalaine Andaya	Project Manager	LogiTeh	lcandaya2@student.apc.edu.ph
Project Stakeholders	See Stakeholder Register	See Stakeholder Register	See Stakeholder Register	See Stakeholder Register
Lead Full-Stack Developer	Atasha Frances Gayle Doria	Lead Developer	LogiTeh	asdoria@student.apc.edu.ph
Full-Stack Developer	Feliz Angelica Salting	Release Manager / Team Lead	LogiTeh	fpsalting@student.apc.edu.ph

Quality Assurance	Michelle Reina Pineda	Quality Assurance	LogiTeh	mbpineda@student.apc.edu.ph
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Communication Methods and Technologies

The LogiTeh project team will determine, in accordance with the Asia Pacific College School of Computing and Information Technologies (APC-SOCIT) guidelines, the communication methods and technologies based on several factors. These include stakeholder communication requirements, available technologies (internal and external), and academic organizational standards.

LogiTeh utilizes an OpenProject Team Site as the primary project management information system. All project team members use this platform to provide updates, track sprint progress, and manage the product backlog. This platform enables the Project Manager and the Project Adviser to monitor project data, activities, and task completion at any point in time.

For the Steering Committee, Defense Panel, and formal document submissions, the team utilizes a dedicated Project SharePoint site. The Project Manager is responsible for ensuring all formal project communications, final deliverables, and documentation are uploaded and organized within this SharePoint platform.

For external stakeholders (such as pilot grooming service providers) who do not have access to OpenProject or the academic SharePoint, communication will be conducted via direct email and virtual meetings through Google Meet.

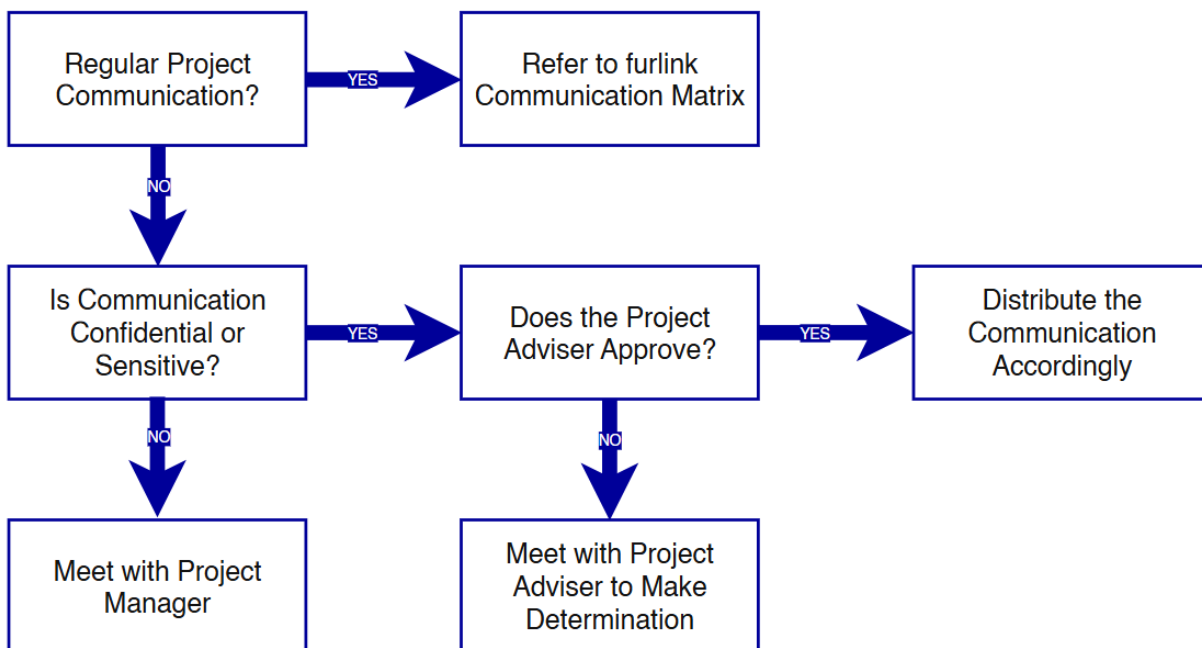
All technical project communication regarding software development and version control, will be maintained and archived on both the GitHub APC-SOCIT project repository and a public GitHub repository of the project.

Communication Type	Objective of Communication	Medium	Frequency	Audience	Owner	Deliverables	Format
Kickoff Meeting	Re-align the team on objectives, schedule, and new requirements.	• Face to Face • Online meeting	Once	• Project Team • Project Adviser	Project Manager	• Agenda • Meeting Minutes	Soft copy archived on PROJMAN team channel
Sprint Planning	Define sprint goals, review product backlog, and assign OpenProject work packages.	• Online meeting	Once during the start of each Sprint	• Project Team	• Project Manager • Scrum Master	• Sprint Backlpg • Updated Sprint Board	OpenProject Dashboard
Daily Stand-Up	Quick sync to discuss what was done yesterday, what will be done today, and identify blockers.	• Online meeting • Brief face to face	Minimum of 2 meetings a week for 15 minutes	• Project Team	• Project Manager • Scrum Master	• Daily Status Update • Identified Blockers	Meeting recording in Microsoft PROJMAN team channel Verbal Updates

Sprint Retrospective	Review what went well during the sprint, identify what didn't, and identify action items to improve next Sprint.	• Online meeting	Once during the end of each Sprint	• Project Team	• Project Manager • Scrum Master	• Retrospective Notes • List of Action Items	Soft copy archived on PROJMAN team channel
Adviser Consultations	Report on project milestones and receive academic guidance.	• Face to face • Online meeting	Once per week	• Project Team • Project Adviser	Project Manager	• Live Demo • Documentation • Meeting Notes	Soft copy of meeting minutes on private channel and PROJMAN team channel
Platform Onboarding and Training	Guide the service provider through using the furlink platform.	• Online meeting • Face to face	Once	• Initial Onboarded Service Providers	Project Manager	• User Manual Guide	PDF Manual, soft copy archived in private team channel and SharePoint

Communication Flowchart

The communication flowchart below was created to aid in project communication. This flowchart provides a framework for the project team to follow for this project. However, there may be occasions or situations which fall outside of the communication flowchart where additional clarification is necessary. In these situations, the Project Manager is responsible for discussing communication with the Project Sponsor and making a determination on how to proceed.



Guidelines for Meetings

Meeting Agenda

For meetings that are not the Sprint Planning, Daily Stand-Ups, and Sprint Retrospective, the meeting agendas will be distributed upon sending a calendar meeting request. The agenda should identify the main presented along with the maximum time for the meeting.

Meeting Minutes

Meeting minutes will be uploaded to the PROJMAN LogiTeh group channel within 2 business days following the meeting. Meeting minutes must follow the meeting minutes template uploaded in the group channel files and must contain action items.

Action Items

Action Items are recorded in the meeting minutes. Action items will include the action item along with who must do the action item. Meetings will start with a review of the status of all action items from previous meetings and end with a review of all action items.

Meeting Chairperson

The Chairperson is responsible for distributing the meeting agenda, facilitating the meeting and ensuring that the meeting minutes are readily available in the team's Team Channel. The Chairperson will ensure that the meeting starts and ends on time and that all presenters adhere to their allocated time frames.

Secretary

The Secretary is responsible for taking the minutes, and helping the facilitator adhere to the time limits set in the meeting agenda. The Secretary will let the presenter know when they are approaching the end of their allocated time.

Parking Lot

The Parking Lot is used by the facilitator to record and defer items which are not on the meeting agenda, however, it will be further discussed at a later time or through another forum. A parking lot record should identify an owner for the item as that person will be responsible for ensuring follow-up.

Communication Standards

For this project, the LogiTech team will utilize standard Asia Pacific College (APC-SOCIT) formats and internal LogiTech templates for all formal project communications. These communications, as detailed in the project's communication matrix, include:

- **Kickoff and Panel Defense:** The project will utilize standard APC-SOCIT presentation templates. Formal meeting minutes and agendas will be drafted using LogiTech document templates and archived on SharePoint and team channels.
- **Sprint Ceremonies:** The project team will utilize standard OpenProject ticket formats to track tasks. Daily stand-up updates will follow a standardized "Yesterday, Today, Blockers" reporting format in meetings and OpenProject Forum.
- **Adviser Status Consultation:** The team will use a standard Project Status Report to summarize milestone progress, budget health, and roadblocks prior to each meeting with the Project Adviser.

Informal project communication must remain professional and effective, but there is no template that must be used outside of the agreed-upon platforms.

Communication Escalation Process

Efficient and timely communication is key to successfully completing furlink by October 2026. Therefore, it is imperative that any disputes, conflicts, or discrepancies regarding project communications are resolved in a way that maintains the project schedule, ensures correct data distribution, and aligns with academic requirements.

To ensure the project stays on track, the LogiTech team will use the following escalation model. This provides a clear hierarchy for resolving communication issues, defining priority levels, decision authorities, and required timeframes for resolution.

Priority Level	Definition	Decision Authority	Timeframe for Resolution
Priority 1: Critical	Major disputes or issues that affect the project's viability, ethical compliance, or funding.	Project Adviser and Steering Committee	Addressed immediately, within 5 hours
Priority 2: High	Conflicts involving project scope changes, academic grading and requirements, or external customer communication.	Project Adviser	Within 3 to 5 business days
Priority 3: Medium	Technical disputes regarding system architecture, GitHub merges, or API integration that may impact the sprint schedule.	Technical Lead and Project Manager	Within 48 hours
Priority 4: Low	Internal miscommunications regarding tasks, OpenProject tickets, or daily sprint goals and blockers.	Project Manager and Team Lead	Within 24 hours

**** NOTE:** Any communication including sensitive and/or confidential information will require escalation to Project Adviser for approval prior to external distribution.

Glossary of Communication Terminology

Term	Definition
Communication	The effective sending and receiving of information. Ideally, the information received should match the information sent. It is the responsibility of the sender to ensure this takes place.
Stakeholder	Individuals or groups involved in the project or whose interests may be affected by the project's execution or outcome.
Communications Management Plan	Portion of the overall Project Management Plan which details how project communications will be conducted, who will participate in communications, frequency of communications, and methods of communications.
Escalation	The process which details how conflicts and issues will be passed up the management chain for resolution as well as the timeframe to achieve resolution.

Scope Management Plan

Introduction

Scope Management is the collection of processes which ensure that the project includes all the work required to complete it while excluding all work which is not necessary to complete it. The Scope Management Plan details how the furlink project scope will be defined, developed, and verified. It clearly defines who is responsible for managing the project's scope and acts as a guide for managing and controlling the scope.

This project is for designing, programming, and testing furlink, a new web-based pet grooming booking and management system tailored for dog and cat owners, as well as local small to medium-sized pet grooming shops. This includes the design of the software, integration of an AI booking assistant, and rigorous testing/validation of the software.

Scope Management Approach

The scope for furlink is defined by the Scope Statement and Work Breakdown Structure (WBS)

The Project Manager, Quality Assurance (QA) Lead, Development team, Project Adviser, and Stakeholders will establish and approve documentation for measuring project scope, which heavily relies on the defined testing phases.

Proposed scope changes may be initiated by the Project Manager, Stakeholders, or any member of the LogiTeh project team. All change requests will be submitted to the Project Manager, who will evaluate the request. Upon acceptance, the Project Manager will submit the scope change request to the Project Adviser for final approval. The Project Adviser is responsible for the acceptance of the final project deliverables and project scope.

Roles and Responsibilities

The table below defines the roles and responsibilities for the scope management of this project.

Role	Name	Responsibilities	Organization/Department
Project Adviser	Gonzalo Gumogda	• Evaluate the necessity and academic alignment of proposed scope changes. • Formally accept and sign off on project deliverables (MVP, v2.0) upon successful scope verification.	APC-SOCIT
Project Manager	Lalainne Andaya	• Maintain overall responsibility for scope management and preventing scope creep. • Measure and verify project scope throughout development. • Facilitate scope change requests, conduct impact assessments, and update project documents upon approval.	LogiTeh
Lead Full stack Developer	Atasha Frances Gayle Doria	• Validate the technical feasibility, infrastructure	LogiTeh

		impact, and schedule impact of any scope change requests. • Ensure all coding and system architecture align strictly with the defined product scope and requirements.	
Full stack developer	Feliz Angelica Salting	• Align approved scope changes with release schedules and deployment cycles. • Ensure that staged deployments only include verified, in-scope features. • Participate in evaluating the impact of scope changes on the sprint backlog.	LogiTeh
QA Lead	Michelle Reina Pineda	• Verify project scope by ensuring deliverables meet the Product Acceptance Criteria. • Facilitate Unit, Integration, System, and UAT phases to confirm all developed features fall within the approved scope.	LogiTeh

Scope Definition

The scope for furlink was defined through a comprehensive requirements collection process and market research. First, a quantitative descriptive survey was conducted involving 64 respondents (43 of which were pet owners from Makati City). Additionally, online interviews were held with owners of local grooming shops to identify specific pain points such as scheduling conflicts, manual tracking inefficiencies, and miscommunication regarding haircut styles.

Project Scope Statement

Product Scope Description

furlink is a centralized digital platform to replace manual pet grooming booking with an efficient platform for pet owners and service providers. By August 2026, the platform aims to achieve four primary strategic objectives: deploying an AI-powered haircut preview tool with 70% rendering accuracy, eliminating booking conflicts via a Gemini 2.5 Flash AI Assistant, onboarding at least one verified provider to improve market visibility, and delivering actionable business insights through a dedicated analytics dashboard. Built on a React.js frontend hosted on Vercel, the system architecture uses Supabase's PostgreSQL engine as its secure backend. Furlink ensures a reliable experience by using Supabase Edge Functions to securely manage AI tasks, automated notifications, and payments in the background, keeping all user data synchronized and secure in real time.

Product Acceptance Criteria

The project will be accepted as complete when it successfully passes the following testing milestones overseen by the QA Lead:

1. Unit Testing: Entry: Individual functions/modules developed and peer-reviewed. Exit: 100% of unit test cases pass, no critical/major bugs, code is at least 80% complete.
2. Integration Testing: Entry: Key modules pass unit testing.
3. User Acceptance Testing (UAT): Entry: System is complete, stable, and ready for stakeholder testing.

Project Deliverables

- furlink v1.0 Optimization
- furlink v1.1 Enhanced Stability Release
- furlink v2.0 Release
- Solution Release furlink v2.1.
- PMBOK Documents

Project Exclusions

It excludes veterinary care, pet insurance, and pet transportation services. Furthermore, the system will not collect or store raw credit card details; all financial handovers are processed via external secure payment links.

Project Constraints

The project budget is constrained to ₱420,374.26 . The project timeline is strictly bound by the academic schedule (April 2026 – October 2026). The system must adhere to the Data Privacy Act of 2012 and must create fallbacks for Gemini AI rate limits and serverless cold-start delays.

Project Assumptions

Target users have internet access and are open to digital solutions and at least one local grooming shop will fully onboard for the MVP launch. Lastly, team members will fully fund the required tech stack.

WBS

To view the work breakdown structure in OpenProject, access this link: [OpenProject WBS](#)

Alternative Analysis

The following alternative options were considered but ultimately not selected for addressing the business problem.

Alternative Option	Reasons For Not Selecting Alternative
No Project (Status Quo)	• Continued high rates of double bookings and scheduling errors. • Lost revenue due to no-shows and lack of enforced liability waivers.
Use Existing Platform: PetBacker	• Unclear business processes; forces users to input detailed pet data before allowing them to explore the platform. • Lack of upfront pricing transparency makes it difficult for owners to compare services within a budget.
Use Existing Platform: Rover	• Geographically restricted; only available and tailored for users in the USA and Canada.
Use Existing Software: DaySmart or Pet Express	• DaySmart: Requires expensive monthly subscription tiers and acts only as an internal tool, offering no public marketplace visibility.

	• Pet Express: A closed, proprietary system exclusive to Pet Express branches, offering no solution for independent local businesses.
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Work Breakdown Structure

APC_2025_2026_T1_MI231_G0... / [Work packages](#) / Favorite: All Work Packages

[← Work packages](#)

All Work Packages

ID	SUBJECT ↑	TYPE	STATUS	ASSIGNEE
5498	▼ 1.0 Initiation Phase	PHASE	New	-
5500	▼ 1.1 PM Foundations Chapter 1: Initiat...	FEATURE	New	-
5503	▼ 1.1.1 Create Project Charter	USER STORY	Closed	FS Feliz Angelica Salting
5623	1.1.1.1 State the scope	TASK	Closed	FS Feliz Angelica Salting
5624	1.1.1.2 Provide summary milest...	TASK	Closed	FS Feliz Angelica Salting
5504	▼ 1.1.2 Create Business Case	USER STORY	Closed	FS Feliz Angelica Salting
5625	1.1.2.1 Provide justification for t...	TASK	Closed	FS Feliz Angelica Salting
5840	1.1.2.2 Update Cost Benefit Ana...	TASK	Closed	FS Feliz Angelica Salting
5626	1.1.2.3 Described the organizati...	TASK	Closed	FS Feliz Angelica Salting
5506	▼ 1.1.4 Create Stakeholder Managem...	USER STORY	Closed	AD Atasha Frances Gayle Dor
5837	1.1.4.1 Add a legend to the mat...	TASK	Closed	AD Atasha Frances Gayle Dor
5628	1.1.4.1 Identify Stakeholders	TASK	Closed	AD Atasha Frances Gayle Dor
5629	1.1.4.2 Create a stakeholder an...	TASK	Closed	AD Atasha Frances Gayle Dor
5507	▼ 1.1.5 Create Stakeholder Analysis	USER STORY	Closed	AD Atasha Frances Gayle Dor
5630	1.1.5.1 Detailed view of the stak...	TASK	Closed	AD Atasha Frances Gayle Dor
5505	▼ 1.2.3 Create Project Brief	USER STORY	Closed	MP Michelle Pineda
5627	1.2.3.1 Provided the high-level ...	TASK	Closed	MP Michelle Pineda
5508	▼ 1.2.6 Create Statement of Work	USER STORY	Closed	LA Lalainne Andaya
5631	1.2.6.1 Provide the scope of wo...	TASK	Closed	LA Lalainne Andaya
5833	1.2.6.1 Update milestone to re...	TASK	Closed	LA Lalainne Andaya

5842	1.2.6.2 Add training of end user...	TASK	Closed	LA	Lalainne Andaya
5632	1.2.6.2 Define time period of th...	TASK	Closed	LA	Lalainne Andaya
5633	1.2.6.3 Describe the actual tasks...	TASK	Closed	LA	Lalainne Andaya
5634	1.2.6.4 Define acceptance criteria	TASK	Closed	LA	Lalainne Andaya
5118	▼ 2.0 Planning Phase	PHASE	New	-	
5621	▼ 2.1 PM Foundations Chapter 2: Plan a ...	FEATURE	In progress	-	
5119	▼ 2.1.1 Plan Communication Manage...	USER STORY	Closed	LA	Lalainne Andaya
5635	2.1.1.1 Define project limitation...	TASK	Closed	LA	Lalainne Andaya
5636	2.1.1.2 Determine the communi...	TASK	Closed	LA	Lalainne Andaya
5637	2.1.1.3 Define roles	TASK	Closed	LA	Lalainne Andaya
5638	2.1.1.4 Determine communicati...	TASK	Closed	LA	Lalainne Andaya
5639	2.1.1.5 Define communication s...	TASK	Closed	LA	Lalainne Andaya
5120	▼ 2.1.2 Plan Scope Management	USER STORY	Closed	FS	Feliz Angelica Salting
5122	2.1.2.1 Collect Requirements	TASK	Closed	FS	Feliz Angelica Salting
5121	2.1.2.1 Plan Scope Management	TASK	Closed	-	
5123	2.1.2.2 Define Scope	TASK	Closed	FS	Feliz Angelica Salting
5124	▼ 2.1.3 Create Work Breakdown Stru...	USER STORY	Closed	FS	Feliz Angelica Salting
6356	2.1.3.1 Confirm the final deliver...	TASK	Closed	FS	Feliz Angelica Salting
5641	2.1.3.2 Organize work packages...	TASK	Closed	FS	Feliz Angelica Salting
5640	2.1.3.3 Organize work packages...	TASK	Closed	FS	Feliz Angelica Salting
5125	▼ 2.1.4 Plan Schedule Management	USER STORY	Closed	FS	Feliz Angelica Salting
5642	2.1.4.1 Provide framework for p...	TASK	Closed	FS	Feliz Angelica Salting
5643	2.1.4.2 Define how the project's...	TASK	Closed	FS	Feliz Angelica Salting
5644	2.1.4.3 Set boundary conditions	TASK	Closed	FS	Feliz Angelica Salting
5645	2.1.4.4 Scope change	TASK	Closed	FS	Feliz Angelica Salting

5126	▼ 2.1.5 Plan Quality Management	USER STORY	Closed	MP	Michelle Pineda
5646	2.1.5.1 Define approach to qual...	TASK	Closed	MP	Michelle Pineda
5647	2.1.5.2 Identify quality requirem...	TASK	Closed	MP	Michelle Pineda
5648	2.1.5.3 Define the process for a...	TASK	Closed	MP	Michelle Pineda
5649	2.1.5.4 Define the process for m...	TASK	Closed	MP	Michelle Pineda
5127	▼ 2.1.6 Plan Cost Management	USER STORY	Closed	LA	Lalainne Andaya
5128	2.1.6.1 Estimate Costs	TASK	Closed	LA	Lalainne Andaya
6355	2.1.6.1 Review costing (for all th...	TASK	Closed	-	
5129	2.1.6.2 Determine Budgets	TASK	Closed	LA	Lalainne Andaya
5130	▼ 2.1.7 Plan Human Resource	USER STORY	Closed	MP	Michelle Pineda
5650	2.1.7.1 Define roles and respon...	TASK	Closed	MP	Michelle Pineda
5651	2.1.7.2 Provide display of projec...	TASK	Closed	MP	Michelle Pineda
5652	2.1.7.3 Note when and how HR ...	TASK	Closed	MP	Michelle Pineda
5663	▼ 2.2 Agile and Project Documents	FEATURE	In progress	-	
5132	▼ 6.1 Create Agile Team Charter	USER STORY	Closed	LA	Lalainne Andaya
5664	6.1.1 Provide specific informati...	TASK	Closed	LA	Lalainne Andaya
5133	▼ 6.2 Create Agile Team Values	USER STORY	Closed	MP	Michelle Pineda
5665	6.2.1 Collaborate with the grou...	TASK	Closed	MP	Michelle Pineda
5135	▼ 7.1 Create System Design Docume...	USER STORY	Closed	AD	Atasha Frances Gayle Dor
5666	7.1.1 Provide System Overview	TASK	Closed	AD	Atasha Frances Gayle Dor
5667	7.1.2 Describe design constraints	TASK	Closed	AD	Atasha Frances Gayle Dor
5668	7.1.3 Provide project reference l...	TASK	Closed	AD	Atasha Frances Gayle Dor
5669	7.1.4 Describe system architect...	TASK	Closed	AD	Atasha Frances Gayle Dor
5333	▼ 3.0 Executing Phase	PHASE	New	-	
5383	▼ 3.2 EPIC 02: Manage Pet Grooming Bo...	EPIC	New	-	
5399	▼ 3.2.4 User Story 04: Submit Feedba...	USER STORY	New	-	
5400	▼ 3.2.4.1 Feature 01: Rating and R...	FEATURE	New	-	
4256	3.2.4.1.2 User prompt for rat...	TASK	On hold	-	
4257	3.2.4.2 AI for pet owner review ...	FEATURE	On hold	-	

[← Work packages](#) All Work Packages

ID	SUBJECT ↑	TYPE	STATUS	ASSIGNEE
5498	▼ 1.0 Initiation Phase	PHASE	New	-
5500	▼ 1.1 PM Foundations Chapter 1: Initiat...	FEATURE	New	-
6671	▼ 1.1.7 Create Business Model Canvas	USER STORY	Closed	FS Feliz Angelica Salting
6674	1.1.7.1 Identify Customer Segm...	TASK	Closed	FS Feliz Angelica Salting
6675	1.1.7.2 Identify Key Partners, M...	TASK	Closed	FS Feliz Angelica Salting
6676	1.1.7.3 Determine how the prod...	TASK	Closed	FS Feliz Angelica Salting
6677	1.1.7.4 Check financial viability	TASK	Closed	FS Feliz Angelica Salting
6673	▼ 1.1.8 Create Feasibility study	USER STORY	Closed	AD Atasha Frances Gayle
6679	1.1.8.1 Technology & Infrastruct...	TASK	Closed	AD Atasha Frances Gayle
6680	1.1.8.2 Marketing & Sales Strat...	TASK	Closed	AD Atasha Frances Gayle
6681	1.1.8.3 HR & Organizational Pla...	TASK	Closed	AD Atasha Frances Gayle
6682	1.1.8.4 Risk Analysis	TASK	Closed	AD Atasha Frances Gayle
5118	▼ 2.0 Planning Phase	PHASE	New	-
5621	▼ 2.1 PM Foundations Chapter 2: Plan a ...	FEATURE	In progress	-
6688	▼ 2.1.10 Create Risk register	USER STORY	Closed	MP Michelle Pineda
6773	2.1.10.1 Standardize Risk Syntax...	TASK	Closed	MP Michelle Pineda
6774	2.1.10.2 consult with devs for sc...	TASK	Closed	MP Michelle Pineda
6689	▼ 2.1.11 Create Configuration manag...	USER STORY	Closed	AD Atasha Frances Gayle
6775	2.1.11.1 Set up the CMDB and a...	TASK	Closed	AD Atasha Frances Gayle
6776	2.1.11.2 Relationship Dependence...	TASK	Closed	AD Atasha Frances Gayle

6690	▼ 2.1.12 Create change management...	USER STORY	Closed	LA	Lalainne Andaya
6778	2.1.12.2 Create impact analysis	TASK	Closed	LA	Lalainne Andaya
6779	2.1.12.3 CCB coordination	TASK	Closed	LA	Lalainne Andaya
6683	▼ 2.1.8 Create Project Risk Managem...	USER STORY	Closed	LA	Lalainne Andaya
6684	2.1.8.1 Create Risk Identificatio...	TASK	Closed	LA	Lalainne Andaya
6685	2.1.8.2 Map the Risk Qualificati...	TASK	Closed	LA	Lalainne Andaya
6686	2.1.8.3 Contingency Planning	TASK	Closed	LA	Lalainne Andaya
6687	▼ 2.1.9 Create Requirements Manage...	USER STORY	Closed	MP	Michelle Pineda
6769	2.1.9.1 Identify the requirements	TASK	Closed	MP	Michelle Pineda
6770	2.1.9.2 define requirements acc...	TASK	Closed	MP	Michelle Pineda
6771	2.1.9.3 Draft Requirements Trac...	TASK	Closed	MP	Michelle Pineda
6772	2.1.9.4 Traceability Testing	TASK	Closed	MP	Michelle Pineda
5484	▼ 4.0 Monitoring and Controlling	PHASE	Closed	-	
6691	▼ 4.1 Chapter 4 PM Foundations monito...	FEATURE	Closed	-	
6692	▼ 4.1.1 Create Project monitoring an...	USER STORY	Closed	FS	Feliz Angelica Salting
6780	4.1.1.1 define the RAG criteria, ...	TASK	Closed	FS	Feliz Angelica Salting
6781	4.1.1.2 Track Weekly/Monthly St...	TASK	Closed	FS	Feliz Angelica Salting
5485	▼ 5.0 Closing Phase	PHASE	New	-	
5486	▼ 5.1 Chapter 5 PM foundations closing ...	SUMMARY TASK	New	-	
5487	▼ 5.1.1 Create transition out plan	USER STORY	New	MP	Michelle Pineda
6782	5.1.1.1 setup strategy for transit...	TASK	Closed	MP	Michelle Pineda
6783	5.1.1.2 knowledge transfer and ...	TASK	Closed	MP	Michelle Pineda
6784	5.1.1.3 Technical and Property ...	TASK	Closed	MP	Michelle Pineda
6785	5.1.1.4 Formal Handover and A...	TASK	Closed	MP	Michelle Pineda
6694	▼ 5.1.2 Create Project acceptance	USER STORY	Closed	AD	Atasha Frances Gayle
6786	5.1.2.1 cross reference RTM and...	TASK	Closed	AD	Atasha Frances Gayle
6787	5.1.2.2 Operational Handover C...	TASK	Closed	AD	Atasha Frances Gayle
6788	5.1.2.3 Administrative Closeout	TASK	Closed	AD	Atasha Frances Gayle
6789	5.1.2.4 Resource Release and Ar...	TASK	Closed	AD	Atasha Frances Gayle
6695	▼ 5.1.3 Create post project review	USER STORY	Closed	LA	Lalainne Andaya
6790	5.1.3.1 Compare the final Actua...	TASK	Closed	LA	Lalainne Andaya
6791	5.1.3.2 perform RCA on any pro...	TASK	Closed	LA	Lalainne Andaya
6696	▼ 5.1.4 Create lessons learned	USER STORY	Closed	FS	Feliz Angelica Salting
6793	5.1.4.1 Analyze both realized an...	TASK	Closed	FS	Feliz Angelica Salting
6794	5.1.4.2 Analyze gaps in current ...	TASK	Closed	FS	Feliz Angelica Salting

Schedule Management Plan

Introduction

The project schedule is the roadmap for how the furlink project will be executed. Schedules are an important part of any project as they provide the project team, project adviser, and stakeholders a picture of the project’s status at any given time.

The purpose of the schedule management plan is to define the approach the LogiTeh team will use in creating the furlink project schedule. furlink is designed to replace manual pet grooming booking processes with an integrated digital ecosystem, centralizing pet registration, scheduling, and payment processing. Because furlink relies on external integrations such as Supabase, PayMongo, Google Gemini AI, and operates within a strict Project-Based Learning (PBL) timeframe, managing the schedule rigorously is critical. This plan includes how the team will monitor the project schedule and manage changes after the baseline schedule has been approved.

Schedule Management Approach

Project schedules will be created using an Agile, sprint-based approach, mapping deliverables from the project’s Work Breakdown Structure (WBS) to specific deployment cycles. Activity sequencing will be highly dependent on the system architecture, prioritizing the React.js frontend, Supabase database configuration, and critical payment and AI integrations.

Duration estimating will factor in known design constraints, such as the time required to configure Supabase Row Level Security (RLS) policies, test PayMongo webhook dependencies, and mitigate Gemini 2.5 Flash AI rate limits. Once a preliminary sprint schedule is developed, it will be reviewed by the project team. Once agreed upon, the Project Adviser will review and approve the schedule before it is baselined.

The following are the designated milestones for the Furlink project schedule:

Milestones	Date
Capstone 2 Kickoff	March 7, 2026
Final Testing	July 6-7, 2026
Release to Service Provider (Training)	July 8, 2026
furlink v1.0 Deployment	July 11, 2026
Enhancement of Core Booking Modules	July 13-17, 2026
Development of AI Booking Assistant	July 20-30, 2026
System and Integration Testing	July 31, 2026 – August 7, 2026
furlink v2.0 Release	August 10 2026
User Acceptance Testing & Fixes	August 17, 2026 – August 21, 2026

Solution Release furlink v2.1	August 31, 2026
Project Completion	October 1, 2026

Roles and responsibilities for schedule development are as follows:

- Project Manager (Lalaine Andaya): Responsible for facilitating work package definition, sequencing, and estimating durations with the team. Creates the project schedule, monitors sprint progress, and obtains schedule approval from the Project Adviser.
- Full-Stack Developer (Feliz Angelica Salting): Responsible for aligning the schedule with the GitHub and Vercel continuous deployment pipeline, ensuring that releases (v1.0, v1.1, v2.0) hit their target deployment dates.
- Lead Developer (Atasha Frances Gayle Doria): Responsible for estimating technical durations, particularly regarding Supabase Functions, and database triggers,
- Lead QA (Michelle Reina Pineda): Responsible for Integration, System, and User Acceptance Testing phases.
- Project Adviser / Sponsor: Participates in reviews of the proposed schedule and approves the final baseline, ensuring it meets Capstone requirements.

Schedule Control

- The project schedule will be reviewed and updated on a bi-weekly basis (aligned with Agile sprint reviews) with actual start, actual finish, and completion percentages provided by the task owners.
- The Project Manager is responsible for determining the impacts of schedule variances, submitting schedule change requests, and reporting status.
- The Project Team is responsible for participating in bi-weekly schedule updates, communicating any technical blockers to the Project Manager, and participating in variance resolution.
- The Project Adviser / Sponsor will maintain awareness of the project schedule status and review/approve any formal schedule change requests submitted by the Project Manager.

Schedule Changes and Thresholds

As the project schedule is created, boundary conditions are established to define the parameters within which the project must operate. Given the strict academic deadlines of the project, timeline adherence is critical. Any event that may potentially cause a schedule

change exceeding these boundary conditions must have a schedule change request submitted and approved.

If any member of the project team determines that a change to the schedule is necessary, the project manager and team will meet to evaluate the change. They will determine impacted tasks, variance severity, and workarounds.

To prevent operational bottlenecks, schedule changes are classified into two tiers:

1. **Minor Schedule Changes:** Low-impact adjustments that do not alter the project baseline or Sprint Goal. The Project Manager and Lead Developer can mutually approve and implement these directly into the current Sprint Backlog.
2. **Major Schedule Changes:** Significant adjustments that alter the project baseline. These must be logged as a formal Change Request and follow a strict approval process.

Once the change request has been reviewed and approved, the Project Manager is responsible for adjusting the schedule, communicating all changes to the team, and archiving the change request in the project repository.

Scope Change

Occasionally, approved changes to the furlink project's scope may result in the schedule needing to be adjusted. These scope changes may arise from technical constraints discovered during development, such as:

- Needing changes to Data Privacy Act compliance features.
- Adjustments to the AI Booking Assistant if the Gemini 2.5 Flash model hits strict rate limiting or context window bottlenecks.
- Modifications to the Role-Based Access Control (RBAC) requiring more backend development time.
- Unforeseen integration issues when configuring the dual-provider redundancy layer (Pollinations.ai to Hugging Face) for the AI Pet Haircut Generator.

Any changes in the project scope approved by the Project Adviser will require the project team to evaluate the effect on the current schedule. If the Project Manager determines that the scope change will significantly affect the timeline, they will request the schedule be formally re-baselined. The Project Adviser must review and approve this request before the new baseline is adopted.

Quality Management Plan

Introduction

The Quality Management Plan is an integral part of the furlink project management plan. The purpose of this plan is to describe how quality will be managed throughout the software development lifecycle (SDLC). It includes the processes for ensuring quality planning, assurance, and control are conducted so that all stakeholders understand how the platform's reliability will be maintained

- Ensure quality is planned
- Define how quality will be managed
- Define quality assurance activities
- Define quality control activities
- Define acceptable quality standards

Quality Management Approach

Quality must be planned into the furlink project to prevent unnecessary rework, coding waste, and project delays. This project utilizes an integrated approach to define software standards, measure code performance, and continuously improve the user experience.

- **Product Quality:** Focuses on the web application's deliverables, ensuring the booking system and user profiles meet customer satisfaction and security standards.
- **Process Quality:** Focuses on the development processes (e.g., Agile sprints, Git workflows) to ensure all coding activities conform to organizational standards

Key Metrics:

- **Schedule & Cost:** Adherence to project milestones and resource allocation.
- **Process Performance:** Sprint velocity and bug fix turnaround time.
- **Product Performance:** Page load speeds, database query efficiency, and mobile responsiveness.
- **Customer Satisfaction:** Results from User Acceptance Testing (UAT) with pet owners and groomers.

Quality Requirements/ Standards

Product Quality:

The furlink project team will document quality standards based on modern web development best practices.

- Compliance Benchmark: The product is considered compliant once we achieve ten consecutive test scenarios (e.g., successful booking, profile update, payment simulation) that result in 100% success within acceptable error margins.

Process Quality:

Process standards will be based on established SDLC frameworks.

- Compliance Benchmark: Process compliance is achieved once all code reviews are completed and deployment metrics fall within acceptable QA margins.

Quality Assurance

Quality assurance for furlink focuses on the iterative processes used during development. This includes measuring sprint metrics, analyzing bug reports, and continuously improving the codebase.

Process Action	Acceptable Standards	Process	Process Phase	Assessment Interval
Code Review	< 5 "Major" bugs found per module		Development	Per Feature/Sprint
API Integration	< 200ms response time from Supabase		Integration	Weekly or per run
UAT Sessions	> 90% positive "Ease of Use" rating		Testing Phase	Per UAT session

The Project Manager will schedule regular reviews to discuss audit findings and process improvement initiatives.

Quality Control

Quality control applies to the furlink product itself. It includes monitoring the results of specific quality activities to assess performance.

Product	Performance Standards	Quality Assessment Activities	Assessment Intervals
Booking Engine	100% data accuracy in Supabase	Automated Unit Testing	Per code commit
Frontend UI	100% Responsive on iOS/Android/Web	Cross-browser testing	Per produced module
Security	SSL Encryption & Secure Auth	Penetration testing	Pre-deployment

Quality Control Measurements

This will be maintained as supporting documentation for final project acceptance.
[LogiTeh_Test Cases.xlsx](#)

Cost Management Plan

Introduction

The Project Manager will be responsible for managing and reporting on the project's cost throughout the duration of the project. During the monthly project status meeting, the Project Manager will meet with major stakeholders to present and review the project's cost performance for the preceding month. Major stakeholders can also view the OpenProject to review and assess the performance and cost of the project. Performance will be measured using earned value. The Project Manager is responsible for accounting for cost deviations and presenting the Project Stakeholder with options for getting the project back on budget. The Project Stakeholder has the authority to make changes to the project to bring it back within budget.

Cost Management Approach

The cost management approach for furlink is designed to optimize resource allocation and maintain fiscal efficiency while ensuring the project remains viable. Given that this is an academic project without a dedicated Project Management Information System (PMIS), costs for this project will be managed at the second level of the Work Breakdown Structure (WBS). Credit for work will be assigned at the work package level on

OpenProject. Progress and resource credit are managed at the work package or user story level. No partial credit is awarded for work-in-progress tasks throughout its development and testing cycle and is only granted 100% of credit upon successfully meeting the team's definition of done and passing quality assurance verification. Financial figures may be rounded to the nearest Philippine peso and work hours rounded to the nearest whole hour.

Cost variances of +/- 0.1 in the Cost Performance Index (CPI) and Schedule Performance Index (SPI) will change the status of the project to cautionary; as such, those values will be changed to yellow in the weekly project status reports. Cost variances of +/- 0.2 in CPI and SPI will change the status of the project to an alert stage; as such, those values will be changed to red in the project status reports. Reaching these alert stages will require immediate action from the Project Manager in order to bring the cost and/or schedule performance indexes below the alert levels. This change must be reviewed and approved by the project stakeholder before it can become within the scope of the project.

While the project encompasses a total calendar duration of six months, the active operational budget accounts for a 3-month billing period. The initial months of the project represented the initiation, requirements gathering, and planning phases, where there are no cloud infrastructure, platform subscriptions, or available API fees were incurred. All development expenses and AI integration costs are calculated strictly for the remaining 3 months of active execution, testing, and deployment to optimize the project's cashflow. Throughout the span of the first three months, software platform and API usages are all strictly free tier.

The budget is segmented into development costs, API costs, and Labor. Where development costs cover platform subscriptions and local operational expenses, API costs for a hybrid strategy where initial development utilizes free-tier models, transitioning to a pay-as-you-go consumption, and Project labor which was calculated at a standardized flat rate across all active working months for all members of the team.

To maintain fiscal conservatism, a standard 10% contingency reserve is applied to the subtotal to absorb unexpected exchange rates, scope changes, or API token overuse.

Measuring Project Cost

Performance of this project will be measured using Earned Value Management. The following four Earned Value metrics will be used to measure project cost performance:

- Schedule Variance (SV)
- Cost Variance (CV)

- Schedule Performance Index (SPI)
- Cost Performance Index (CPI)

If the SPI or CPI has a variance of between 0.1 and 0.2, the Project Manager must report the reason for the exception. If the SPI or CPI has a variance greater than 0.2, the Project Manager must report the reason for the exception and provide a detailed corrective action plan to bring the project's performance back to acceptable levels.

Performance Measure	Yellow	Red
Schedule Performance Index (SPI)	Between 0.9 and 0.8 or Between 1.1 and 1.2	Less Than 0.8 or Greater than 1.2
Cost Performance Index (CPI)	Between 0.9 and 0.8 or Between 1.1 and 1.2	Less Than 0.8 or Greater than 1.2

Reporting Format

Reporting for cost management will be included in the monthly project status report. The Monthly Project Status Report will include a section labeled, "Cost Management". This section will contain the Earned Value Metrics identified in the previous section. All cost variances outside of the thresholds identified in this Cost Management Plan will be reported on including any corrective actions which are planned. Change Requests which are triggered based upon project cost overruns will be identified and tracked in this report.

Cost Variance Response Process

The Control Thresholds for this project is a CPI or SPI of less than 0.8 or greater than 1.2. If the project reaches one of these Control Thresholds a Cost Variance Corrective Action Plan is required. The Project Manager will present the Project Stakeholder with options for corrective actions within five business days from when the cost variance is first reported. Within three business days from when the Project Stakeholder selects a corrective action option, the Project Manager will present the Project Stakeholder with a Cost Variance Corrective Action Plan. The Cost Variance Corrective Action Plan will detail the actions necessary to bring the project back within budget and how the effectiveness of the actions in the plan will be measured. Upon acceptance of the Cost Variance Corrective Action Plan, it will become a part of the project plan, and the project will be updated to reflect the corrective actions.

Cost Change Control Process

The cost change control process will follow the established project change request process. Approvals for project budget/cost changes must be approved by the project sponsor.

Project Budget

The budget for this project is detailed below. Costs for this project are presented in various categories:

Category	Budget
Software and API Costs	₱38,2158.41
Labor Costs	₱360,000.00
Contingency Reserve	₱38,215.84
Total Project Cost	₱ 42,0374.26

Human Resource Plan

Introduction

Human resources management is a critical component of the furlink project. This plan serves as a tool to aid in the management of human resource activities from development through to User Acceptance Testing (UAT) and final deployment Roles and responsibilities of team members throughout the project

The furlink Human Resource Plan includes:

- Roles and Responsibilities: Clear definitions for team members.
- Project Organization Charts: Visualizing hierarchy and task assignments.
- Staffing management plan to include:
 - a. How resources will be acquired
 - b. Timeline for resources/skill sets
 - c. Training required to develop skills
 - d. How performance reviews will be conducted
 - e. Recognition and rewards system

The goal is to ensure the project has the necessary technical skills (React.js, Supabase, QA testing) and that team activities are effectively managed to meet the project's social impact goals.

Roles and Responsibilities

The roles and responsibilities for the Software Upgrade Project are essential to project success. All team members must clearly understand their roles and responsibilities in order to successfully perform their portion of the project. The following project team roles and responsibilities have been established:

Project Manager (PM), (1 position): Responsible for the overall success of the Project. The PM must authorize and approve all project expenditures. The PM is also responsible for approving that work activities meet established acceptability criteria and fall within acceptable variances. The PM will be responsible for reporting project status in accordance with the communications management plan. The PM will evaluate the performance of all project team members. The PM must possess the following skills: leadership/management, budgeting, scheduling, and effective communication.

Lead Full-Stack Developer (LD), (1 position): Accountability for the end-to-end technical architecture, code security, and final deployment of the furlink web application. The LD holds the authority to make critical decisions regarding the core tech stack, external library integrations, database schema modifications, and deployment workflows. Primary responsibilities include designing the system architecture, setting up the Supabase database infrastructure, configuring third-party APIs (Gemini AI and PayMongo), and overseeing the repository's main code branch. This role requires advanced technical competency in Vite, Node.js, and PostgreSQL, along with strong leadership skills to guide code freezes, conduct code reviews, and cross-train team members on database conflict resolution.

Full-Stack Developer (FD), (1 position): Responsible for the core feature development, implementation, and maintenance of both the user interface and backend application logic. The FD has the authority to implement design patterns, write specific API queries, and construct reusable components within the established architectural guidelines. Primary responsibilities include translating user stories into responsive UI components using Vite/CSS, managing Supabase backend data queries, handling CRUD operations for provider listings, and building

real-time dashboard analytics. This role requires technical proficiency in JavaScript, CSS, and database manipulation, with a focus on ensuring the platform functions seamlessly across various mobile viewports and browser environments.

QA Tester (QA): The QA Tester focuses on ensuring the reliability and user-friendliness of the platform by developing comprehensive test plans that cover both manual functional testing and automated regression suites. This role holds the authority to flag critical bugs that could delay a release, ensuring that the software is thoroughly validated before it reaches the end user. Responsibilities include documenting all identified issues, verifying technical fixes, and facilitating User Acceptance Testing (UAT) sessions with actual pet service providers to measure the platform's ease of use. Success in this role requires strong analytical skills and a deep understanding of the Software Testing Life Cycle (STLC) to prevent data errors or UI failures.

Project Organizational Charts

The following RACI chart shows the relationship between project tasks and team members. Any proposed changes to project responsibilities must be reviewed and approved by the project manager. Changes will be proposed in accordance with the project's change control process. As changes are made, all project documents will be updated and redistributed accordingly.

Project Phase / Activity	Project Manager	Lead Full-Stack Dev	Full-Stack Dev	QA Tester
Requirements Gathering	R	A	C	C
Database & Schema Design	A	R	R	I
Frontend & Backend Coding	I	A	R	C
Software & Unit Testing	A	C	C	R
User Acceptance Testing (UAT)	A	I	I	R
Environment Preparation	A	R	C	I
Final Implementation / Release	A	R	C	I

Project Documentation	A	R	R	R
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Key:

R – Responsible for completing the work

A – Accountable for ensuring task completion/sign off

C – Consulted before any decisions are made

I – Informed of when an action/decision has been made

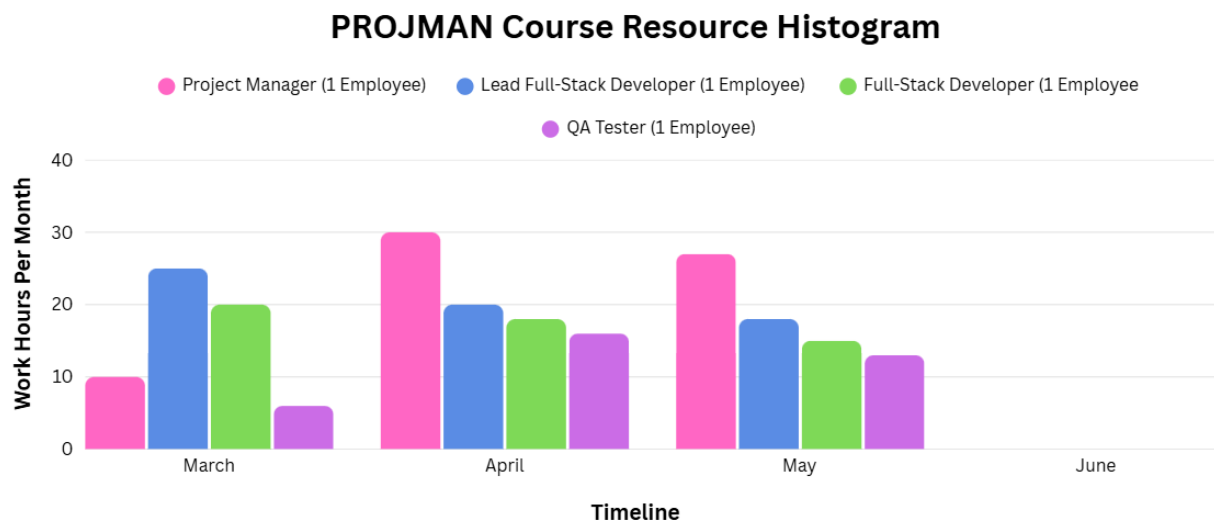
Staffing Management

Staff Acquisition:

Project staff for furlink consists of internal student resources. The Project Manager identifies and assigns resources based on the technical requirements of the platform. All resources will work remotely or from the designated campus workspace.

Resource Calendars:

The PROJMAN Course will last for four months. All resources are required before the project begins. The resource histogram below illustrates that the Project Manager, Full-Stack Developers, and QA Tester will all perform work for a maximum of 40 hours per month from March to June.



Performance Reviews:

The PM will evaluate team members based on the successful completion of "sprints" and the quality of code/documentation produced. Feedback will be provided during project status meetings.

Recognition and Rewards:

To maintain motivation throughout the development of the pet grooming platform:

- **Completion Event:** A "Launch Party" for the team upon the successful first booking on the platform.

Risk Management Plan

Introduction

furlink is an online platform connecting dog and cat owners with pet care service providers. As the project moves toward its Alpha Release and subsequent versions, it operates in an area of uncertainty inherent in developing unique AI-integrated services. This plan establishes the framework for identifying, scoring, and mitigating risks to ensure the project meets its goals within the defined constraints of time, scope, and cost.

The project is currently categorized as High Risk due to several factors with high probability and high impact, such as limited manpower and limited financial resources.

Before risk management begins it is imperative that a foundation established for providing structured project information, thus, the following project elements were completed and defined prior to developing this Risk Management Plan:

- Define work scope, schedule, resources, and cost elements
 - Develop project Work Breakdown Structure
 - Develop master schedule and detailed schedules
 - Estimate project cost and finalize budget
 - Identify required and available resources
 - Establish performance measurement metrics
- Define minimum and maximum baseline thresholds
 - Schedule
 - Resources
 - Cost
- Baseline reporting requirements
 - Format
 - Frequency of distribution
 - Distribution list

- Define Risk Management Roles and Responsibilities
 - Project Manager chairs the risk assessment meetings
 - Project team participates in risk assessment meetings and members serve as meeting recorder and timekeeper
 - Key stakeholders participate in risk assessment meetings
 - Project Sponsor may participate in risk assessment meetings

Top Three Risks

The following risks have been identified as having the highest potential impact on the **furlink** release schedule and functionality:

Low Provider Adoption

Targeted Makati service providers are hesitant to move away from manual booking.

AI API Cost Overrun

The intensive transactional token consumption driven by multi-turn dialogs within the AI Booking Assistant and massive data array parsing by the AI-Powered Data Analyst—coupled with heavy GPU graphic rendering pipelines inside the AI Pet Haircut Generator—exceeds the core team's allocated fixed budget, causing premature deployment throttling, platform downtime, or financial operational deficits.

AI Hallucination Risk

AI features generate incorrect schedules, pricing, or advice, leading to grooming appointment schedules.

Risk Management Approach

The approach taken to manage risks for the **furlink** project involves a methodical, agile-based process integrated directly into the Capstone 2 schedule. The project team continuously logs and evaluates risks, ensuring that high-priority threats are explicitly tracked alongside core development milestones. Status updates on these risks are a mandatory component of bi-weekly project team meetings, specifically led by the Release Manager and QA Lead. Upon the completion of the project in October 2026, the Project Manager will analyze the overall risk management lifecycle to document improvements and lessons learned for future software iterations.

Risk Identification

Risk identification for the furlink platform relies on collaborative brainstorming and historical analysis to uncover potential threats before they impact the deployment schedule. The development team utilizes consultations with the Project Adviser and Co-Adviser to gain strategic foresight into common project pitfalls. Additionally, the Project Manager, Lead Developer, and QA Lead conduct dedicated risk assessment meetings to pinpoint technical challenges specifically related to AI integration.

Risk Qualification and Prioritization

Once potential risks are identified, they are qualified and prioritized to determine their potential impact on the Alpha and v2.0 releases. The team evaluates the probability of each risk occurring against the severity of its impact on core system functionalities and strict academic deadlines using a 5x5 Matrix. Critical risks demand immediate and robust mitigation strategies. Conversely, low-impact issues, such as minor cosmetic UI misalignments, are assigned a lower priority to ensure limited development resources remain strictly focused on essential platform stability.

Risk Monitoring

Active risk monitoring is a continuous, sprint-based process spanning the entire development lifecycle of the furlink platform. As risks approach their anticipated timeframes within the Capstone schedule, the Project Manager is assigned to oversee and report on them. The team utilizes GitHub Issues to dynamically log, track, and monitor technical defects and schedule delays in real-time.

Risk Mitigation and Avoidance

The project team carefully evaluates all identified risks against the strict project constraints of time, scope, and cost to formulate effective mitigation and avoidance strategies. Time and scope are treated as rigid, inflexible constraints; the October 2026 completion date cannot be delayed, and the core MVP functionalities must be delivered for academic grading. If unforeseen technical hurdles threaten the schedule, the team will mitigate the risk by reducing the project's scope, decisively cutting secondary features rather than extending the timeline. Cost remains the only flexible constraint, serving as a

final contingency where team members might allocate personal funds to temporarily upgrade server capacities if user traffic overwhelms the free cloud tiers during User Acceptance Testing.

Requirement Management Plan

Introduction

The purpose of the furlink Requirements Management Plan is to establish a common understanding of how requirements will be identified, analyzed, documented, and managed for the pet grooming platform. The project is currently categorized as High Risk due to several factors with high probability and high impact, such as limited manpower and limited financial resources.

Requirements are divided into two categories:

- Project Requirements: Non-technical needs such as onboarding one (1) service provider in Makati and completing the project by October 2026.
- Product Requirements: Technical specifications for the WebApp, including the AI-Booking Assistant, Supabase backend integration, Business analytics dashboard, and PayMongo payment link automation.

Requirements Management Approach

Identification

Requirements will be collected through interviews with local Makati pet groomers, focus groups with pet owners, and prototyping of the furlink WebApp UI.

Analysis

Requirements will be categorized into "Core Booking Logic," "AI Tools," or "Business Analytics." The team will determine acceptance criteria for each.

Documentation

Analyzed requirements will be added to the project requirements checklist and assigned to the Lead Developer and QA Tester.

Ongoing Management

The Project Manager will track status during sprints. Changes must go through the established change control process to assess impact on the launch date.

Configuration Management

Documentation & Version Control: All technical designs and requirements will be stored in a central repository. Changes to the database schema or logic require PM approval.

Change Control: Any proposed changes to the project scope must be reviewed by the project team to determine the impact on the August 2026, deployment deadline and the overall budget.

Requirements Prioritization Process

furlink uses a three-level scale to ensure the most critical features are delivered first:

High Priority: Essential for launch

- Secure User registration and login
- Booking Engine
- Service listing
- Profile Dashboard
- Payment Gateway

Medium Priority: Important features that add value

- AI Booking Assistant
- Feedback Form

Low Priority: Desirable features for future updates

- Advanced Business Analytics
- Multi-city expansion

Product Metrics

To gauge project success, the following technical and performance metrics are established:

Cost & Performance:

- Operating Cost: Server and database costs (Supabase) must stay within the monthly student budget.
- Speed: PayMongo payment links must be delivered instantly; page load times must be < 2 seconds.

Technical Quality:

- Accuracy: 100% accuracy in understanding user booking requests by the system.

Requirements Traceability Matrix

Project Name	furlink	Business Area	Pet Services			
Project Manager	Lalaine Andaya	Business Analyst Lead	Lalaine Andaya			
QA Lead	Michelle Reina Pineda	Target Implementation Date	October 1, 2026			
Req#	Requirement Description	Design Document Reference	Charter Reference	Test Case Reference	User Acceptance Validation	Comments
PR-01	User account registration.	UC-1.1	3.1.1.1.4	UC-1.1 TC		
PR-02	User login functionality.	UC-1.2	3.1.2.1	UC-1.2 TC		
PR-03	Account management (Profile updates).	UC-1.3, UC-1.4	3.2.3.2.1, 3.1.4.1	UC-1.3 TC, UC-1.4 TC		
PR-13	Switch account from Pet Owner to Service Provider.	UC-1.5	3.1.5.1	UC-1.5 TC		
PR-04	Browse pet services on the platform.	UC-2.1	3.2.1.1.8	UC-2.1 TC		
PR-05	Book an appointment with a service provider.	UC-2.1	3.2.1.1.9	UC-2.1 TC		
PR-15	Cancel existing booking.	UC-2.2	3.2.2.1	UC-2.2 TC		
PR-16	Reschedule booking (prior to provider acceptance).	UC-2.3	3.2.3.1	UC-2.3 TC		
PR-06	Share feedback/reviews for service providers.	UC-2.4	3.2.4.1.1	UC-2.4 TC		
PR-19	Submit payment verification for approved bookings.	UC-2.5	3.2.5.1	UC-2.5 TC		
PR-09	AI Booking Assistant / Haircut Preview generation.	UC-2.6	3.2.6.1	UC-2.6 TC		
PR-07	Create, Read, Update, and Delete (CRUD) listings.	UC-3.1	3.3.1.2	UC-3.1 TC		
PR-08	View posted listings.	UC-3.3	3.3.3.1	UC-3.3 TC		
PR-20	Delete posted listings.	UC-3.4	3.3.4.1	UC-3.4 TC		
PR-23	Modify/Edit existing listings.	UC-3.2	3.3.2.1	UC-3.2 TC		
PR-12	View performance dashboards.	UC-4.1	3.4.1.1.7	UC-4.1 TC		
PR-14	Manage (Accept/Decline) booking requests.	UC-3.5	3.3.5.1	UC-3.5 TC		
PR-17	View and download booking reports.	UC-4.1	3.4.1.1.8	UC-4.1 TC		
PR-21	Confirm or decline user payment verification.	UC-3.6	3.3.6.1.2	UC-3.6 TC		
PR-10	Secure login access to administrative interface.	UC-1.2	3.1.2.2	UC-1.2 TC		
PR-11	Manage and monitor user activities.	UC-5.1	3.5.1.1	UC-5.1 TC		
PR-18	View and download user activity reports.	UC-4.1	3.4.1.1.9	UC-4.1 TC		
PR-22	Accept or reject new service provider listings.	UC-5.2	3.5.2.1	UC-5.2 TC		

Risk Register

Risk Identification				Qualitative Rating			Risk Responses		
Risk Ranking	Risk Title	Risk Description	Risk Category	Probability	Impact	Risk Score	Risk Response	Trigger	Risk Owner
1	Low Provider Adoption	Targeted Makati service providers are hesitant to move away from manual booking.	Business and Reputation	5	5	25	Provide in-person onboarding and "concierge" profile setup services to lower barrier to entry.	Less than 1 verified provider by May 15 release.	Project Manager
2	AI API Cost Overrun	The intensive transactional token consumption driven by multi-turn dialogs within the AI Booking Assistant and massive data array parsing by the AI-Powered Data Analyst exceeds the core team's	AI and Third-Party Dependencies	5	5	25	Implement programmatic server-side middleware rate limiting alongside robust token optimization, semantic compression arrays, and payload pruning rules within the Supabase Edge Functions.	A sudden, unpredicted spike in legitimate user traffic or high-turn conversational chat sessions following your marketing campaigns.	Lead Full-Stack Developer

		allocated fixed budget, causing premature deployment throttling, platform downtime, or financial operational deficits.							
3	AI Hallucination Risk.	AI features generate incorrect schedules, pricing, or advice, leading to grooming appointment schedules	AI and Third-Party Dependencies	3.25	5	16.25	A hidden code guardrail checks all AI answers against the real database and forces the user to review a final summary screen before anything is officially booked.	Confusing prompts or inputs from users.	Lead Full-Stack Developer
4	Unauthorized Admin Access	Malicious actors bypassing security controls to gain full administrative privileges.	Security and Compliance	3	5	15	Implement multi-factor authentication (MFA) for admin panels and strict IP-whitelisting for internal tools.	User feedback flags 2+ automated booking errors in a single day.	Full-Stack Developer
5	Security Breach	Unauthorized access to	Security and	3	5	15	Implement HTTPS, industry-	Any login attempt to	Lead Full-Stack

		sensitive user or payment data.	Compliance				standard hashing, and strict Role-Based Access Control (RBAC).	/admin from an unrecognized IP address.	Developer
6	UAT Failure	Pet owners find the interface confusing, leading to low "Ease of Use" ratings.	Quality	2.75	4.5	12.375	Conduct weekly UI reviews and iterative feedback sessions before the final August UAT. (regular)	Average ease-of-use rating falls below 3.5/5.	Quality Assurance
7	Database Failure	Technical Infrastructure and Performance	Technical Infrastructure and Performance	2	5	10	Set up automated daily backups and point-in-time recovery (PITR) inside Supabase to ensure data can be restored instantly if the system crashes or corrupts.	The application throws continuous database connection errors like timeouts and 500 Internal Server Error.	Lead Full-Stack Developer
8	Dashboard Data Processing	Critical crash of cloud database infrastructure, resulting in temporary loss of data.	Technical Infrastructure and Performance	2.25	3.75	8.4375	Build an API cache layer in the backend to store analytical reports temporarily, and implement a manual	Dashboard metrics fail to refresh, throw loading timeout errors, or display zero/bla	Full-Stack Developer

							refresh button so providers only fetch fresh data when absolutely necessary.	nk values during active business periods.	
9	Data Privacy Compliance Violation	Mishandling user consent or failing to honor user privacy rights, triggering legal penalties under the Philippine Data Privacy Act (R.A. 10173).	Security and Compliance	1.25	5	6.25	Design data registration screens to explicitly capture user consent via checkboxes, and build an automatic one-click "Delete Account" button so users can instantly wipe their records.	Any unauthorized data access attempt detected.	Lead Full-Stack Developer
10	Loss of Customer Trust	Brand damage resulting from recurring technical issues, data breaches, or consistently poor user experience.	Business and Reputation	1.25	4.25	5.3125	Publish real-time status pages; proactively dispatch compensation discounts to affected Makati pet owners.	Official security advisory from the AI provider or detection of anomalous AI outputs.	Product Owner
11	Fraudulent Payment Shutdown	Fraudulent or suspicious	Security and Compliance	1.5	3.5	5.25	Deploy client-side transaction screening	Database connection pool	Project Manager

		transactions may trigger payment gateway security measures, causing temporary account freezes and delayed fund transfers.					logic; hold booking validation until webhook confirms clear payment.	fails or database returns persistent 500 connection drop codes.	
12	Payment Gateway Delay	Issues with PayMongo integration prevent automated payment link delivery.	AI and Third-Party Dependencies	1.5	3.25	4.875	Build an automatic background retry loop into your Supabase Edge Functions that programmatically re-attempts to deliver the payment link if the initial PayMongo connection fails.	The PayMongo API endpoint times out or returns an error status code during payment.	Lead Full-Stack Developer
13	High Traffic Issues	Server overload during peak booking seasons or hours, causing the application to	Technical Infrastructure and Performance	1.5	3	4.5	Configure Vercel auto-scaling thresholds to dynamically spin up serverless computing nodes during traffic spikes,	Real-time CPU or memory utilization on your active cloud infrastructure surges	Lead Full-Stack Developer

		significantly slow down or crash.					implement a Supabase API caching layer to minimize heavy database reads, and set up a lightweight traffic-throttling middleware to queue incoming requests safely.	past 85%	
14	Mobile Device Compatibility Issues	The platform interface breaks, overlaps, or renders improperly on various smartphone screens, rendering key buttons unusable for pet owners trying to book on the go.	Technical Infrastructure and Performance	1.25	3.5	4.375	Enforce mobile-first responsive CSS architecture and conduct automated cross-browser layout testing across standard iOS and Android screen profiles before every code deployment.	The staging test suite reports a responsive breakpoint mismatch, or live user session logs record a sudden drop in booking completions specifically from mobile viewports.	Full-Stack Developer

15	Incomplete System Updates	Deploying flawed or partially finished code to production environment, causing system instability or breaking existing features.	Operational	1	4	4	Establish a strict main-branch protection policy in GitHub requiring automated test passes and mandatory approval signatures from both QA and Release leads before code can be merged into production.	Automated integration tests fail during a deployment run, or a sudden spike in software crash alerts is logged immediately following a new platform update.	Quality Assurance
16	Third-Party Service Shutdown	Sudden discontinuation or prolonged outage of critical external platforms.	AI and Third-Party Dependencies	1	3.25	3.25	Abstract all third-party API dependencies into modular code wrappers, allowing the development team to quickly swap connections and migrate to an alternative provider with minimal code rewrites.	An external vendor issues an official product sunset announcement, or the platform's system logs record a continuous 24+ hour service drop from a core API provider.	Lead Full-Stack Developer

17	Down AI Image Generation	Outage of external AI tools preventing the creation AI generated image.	AI and Third-Party Dependencies	1.5	2	3	Implement an automated fallback system in your code that temporarily routes image requests to a backup AI engine while the primary AI service is offline.	The platform's backend logs a high rate of connection timeouts or 503 Service Unavailable error responses specifically from the primary image generation API.	Full-Stack Developer
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Configuration Management Plan

Introduction

The purpose of the Configuration Management (CM) Plan is to establish and describe the systematic framework for conducting configuration management throughout the entire project lifecycle of furlink. By defining a structured, coordinated approach, this document details how configuration items are tracked, implemented, verified, and communicated to project stakeholders, thereby ensuring that all system builds, database schema trees, and technical documentation remain consistent and accurate. In the absence of a formalized configuration management plan, software and information technology initiatives are highly vulnerable to version conflicts, omitted requirements, or redundant work driven by a lack of strict baseline and change controls. Given that furlink is an advanced digital booking ecosystem designed to replace informal pet care arrangements with a structured, cloud-native marketplace, maintaining structural integrity across parallel development lines is paramount. This necessity is further amplified by the inherent technical complexity of integrating a modern serverless architecture composed of React.js, Supabase, Vercel compute nodes, the PayMongo transactional gateway, Gmail SMTP communication relays, and the Google Gemini 2.5 Flash API. Consequently, this plan acts as the authoritative strategy for preventing architectural divergence across sandbox testing, staging, and production environments, while explicitly establishing the roles of the Configuration Control Board (CCB) and defining the maintenance procedures for the platform's digital assets.

Roles and Responsibilities

The furlink Configuration Control Board (CCB) functions as a unified governing body composed of the entire core project team. Because the platform utilizes a flat startup layout with modular development boundaries, all members are responsible for evaluating the structural and architectural feasibility of proposed infrastructure transformations.

- Lalainne Andaya (Project Manager): The furlink Configuration Control Board (CCB) functions as a unified governing body composed of the entire core project team. Because the platform utilizes a flat startup layout with modular development boundaries, all members are responsible for evaluating the structural and architectural feasibility of proposed infrastructure transformations.

- Atasha Frances Gayle Doria (Lead Full-Stack Developer): Holds ultimate accountability for platform architecture, serverless infrastructure synchronization, and repository integrity. She is responsible for assigning version identifiers, managing branch permissions, configuring environment variables across production hosts, and holding final tie-breaking authority over code-level technical implementation decisions.
- Michelle Reina Pineda (Quality Assurance): Responsible for drafting formal change requests when module telemetry or user feedback requires adjustments to the software baseline. She ensures all locally engineered features comply with repository standards and evaluates the side-effect impacts of stack updates on neighbor dependencies.
- Feliz Angelica Salting (Full-Stack Developer): Formulates drafts comprehensive CCRs when user or provider feedback necessitates adjustments to active feature baselines. Responsible for assessing cross-modular technical impacts during board evaluations, ensuring that approved code alterations strictly follow established repository templates and standards.

Configuration Control

Configuration Control for furlink is the process of systematically evaluating, controlling, and managing all software framework, database schema, and cloud infrastructure modifications throughout the project lifecycle. To prevent processing bottlenecks while maintaining system stability, change request are classified into two strict categories based on their technical impact. Major Changes refer to any modifications that alter the core project scope, timeline, database schema structures, primary API endpoints—such as migrating from Google Gemini to an alternative language engine—or budget parameters. These high-impact requests must strictly undergo the full, formal Change Control Board evaluation loop and require unanimous consensus from the team to proceed. Conversely, Minor Changes encompass low-risk, low-impact development adjustments that do not alter the core architecture, modify database columns, or jeopardize the project baseline. These include simple user interface alignment fixes, text revisions, or minor dependency patch updates. Minor changes bypass formal board routing entirely and can be mutually cleared by the Project Manager and Lead Developer as simple tasks during standard sprint backlogs.

The end-to-end infrastructure change workflow begins with identification, where technical bugs, stack deprecations, or performance shortfalls are captured and

documented within OpenProject as formal tickets. Next, during the Preliminary Analysis phase, the Project Manager reviews the ticket to evaluate its scope and cost impact, formally classifying the submission as either a Major or a Minor Change. If the request is classified as a Minor Change, it bypasses the formal board assessment and moves directly into sprint allocation; however, if it is classified as a Major Change, it is formally scheduled for a board review session. During the Board Evaluation phase, the entire project team reviews the proposal against two critical criteria: its explicit alignment with the platform's market goals and its technical feasibility within the current cloud-native tech stack. Denied requests are logged but may be updated with new data and re-submitted later for board reconsideration.

Once a major change request is approved by the board, the lifecycle transitions into Implementation and Isolation, where the Project Manager re-baselines the project documentation and scope metrics, while the Lead Developer simultaneously authorizes the creation of a dedicated Feature Branch in GitHub to isolate the infrastructure configuration update away from production code. During active development, any resulting technical blockers or configuration anomalies are reported transparently during the team's daily stand-up meetings. Following the completion of code updates, the project undergoes Testing and Verification, an essential quality assurance check where comprehensive software and regression testing are executed within an isolated staging environment, allowing the Quality Assurance to validate that the new infrastructure components remain perfectly stable. Finally, the process reaches Closure once all features are fully verified; the change request ticket is officially closed within OpenProject, and the Lead Full-Stack Developer safely merges the staging codebase into the master development branch for automated, live cloud deployment.

Configuration Management Database (CMDB)

For the furlink platform, the CMDB is structured as a coordinated ecosystem of three specialized cloud tools to maintain a complete history of project assets. First, GitHub acts as the source code and database migration schema registry, providing a complete, auditable history of every code adjustment alongside its author, timestamp, and commit description. Second, Microsoft 365 manages project documentation through a tiered strategy, utilizing Microsoft tools as the primary hub for live document co-authoring and SharePoint as the shared workspace for finalized, Read-Only project artifacts mirrored into the academic repository. Third, Vercel and Supabase coordinate to manage live serverless computing infrastructure, data configurations, and system environment variables. Vercel acts as the host environment variable registry by dynamically injecting encrypted API

credentials—such as the PayMongo authentication secrets, GMAIL SMTP application keys, and Google Gemini connection string tokens—at the cloud edge. Simultaneously, supase serves as the repository for the relational PostgreSQL database schema structures, tracking SQL tables and row-level security policies to allow the Lead Full-Stack Developer to handle swift rollbacks if database discrepancies are identified.

Configuration Status Accounting

Configuration Status Accounting for furlink involves the continuous collection and reporting of the status of all CIs to ensure the team and stakeholders have real-time visibility into the project's progress. To track and account for active infrastructure assets, the team utilizes a combination of digital tracking systems and synchronous sprint checkpoints. OpenProject serves as the primary status accounting database, where individual team members are responsible for updating the status logs of their assigned work packages to accurately reflect its state. Complementing this platform, the team conducts stand-up meetings to synchronize development trajectories and instantly flags any open configuration change requests (CCRs) acting as code or architectural blockers.

To satisfy reporting requirements, the Project Manager compiles and submits comprehensive status summaries during scheduled milestones. These evaluations systematically report on change request metrics, including aging distributions to track how long infrastructure changes remained open, and trending maps to identify which specific architectural modules are undergoing the most frequent code adjustments. Additionally, status accounting reports detail version control metrics across software frames, data structures, and supporting documentation, while providing build logs that detail file states, incorporated changes and active CI relationships. Prior to any formal product release, the Lead Full-Stack Developer works closely with the Project Manager to verify that all system dependencies are updated to their correct release numbers, while preparing formal configuration summaries to be presented to panelists for high-level project validation during scheduled defense windows.

Configuration Audits

Configuration audits will occur as an ongoing pre-release check prior to any major software launch or at the discretion of the Project Manager and project advisers. All configuration audits are performed directly by the Lead Full-Stack Developer in close

coordination with the Project Manager, ensuring that all code alterations and deployment procedures follow the guidelines established in this plan.

The audit workflow requires the execution of six distinct technical evaluation tasks:

1. The Lead Full-Stack Developer establishes an audit environment by spinning up an isolated, temporary clone staging sandbox within Vercel to inspect modifications without altering live system traffic.
2. The Lead Full-Stack Developer copies the latest software builds, database schemas, and documentation files into this audit environment for holistic review.
3. The Lead Full-Stack Developer verifies versioning and numbering conventions, ensuring that all code dependencies in package.json are anchored to immutable version states according to the defined branching strategy
4. The Lead Full-Stack Developer conducts a historical analysis by cross-referencing timestamps and commit descriptions in GitHub and OpenProject to ensure all baseline modifications were properly recorded.
5. The Lead Full-Stack Developer, in coordination with the Quality Assurance, executes testing on the staging build to confirm that all functional requirements are perfectly met.

The Lead Full-Stack Developer conducts a comprehensive artifact review to ensure all system secrets are securely partitioned and that all approved CCRs have been fully incorporated and logged in the system records.

Change Management Plan

Introduction

Change Management is a critical component of the project. Because the platform relies on complex integrations, unforeseen technical challenges or shifting requirements are inevitable. The Change Management Plan for the project establishes clear expectations on how proposed changes will be vetted, managed, and communicated. If changes to the platform's scope, tech stack, or timeline are implemented in an unorganized manner, the project risks missing critical Capstone milestones. All stakeholders, including the

development team and project advisers, are expected to submit and process changes in accordance with the processes detailed in this document.

Change Management Approach

The approach taken to manage change for the furlink project is designed to be highly protective of the core timeline. Because time and scope are rigid constraints dictated by academic deadlines, the primary goal of this approach is to prevent scope creep while allowing necessary technical pivots. The project team will ensure that all proposed changes are rigorously defined and evaluated against the project's ability to hit the completion date. Only changes that are essential to delivering the core MVP will be approved. By maintaining this strict methodology, the development team will focus its limited resources solely on beneficial adjustments, avoiding unnecessary feature enhancements that could jeopardize the final solution release.

Definitions of Change

Several types of changes may be requested and considered during the development of furlink, each carrying different impacts on the overall project plan.

Scope Changes

Any additions or subtractions to the features outlined, such as the AI Booking Assistance's capabilities or altering service provider dashboard metrics.

Technical and Architectural Changes

These refer to the fundamental shifts in the underlying technology stack such as migrating away from Vercel or replacing Supabase with NoSQL solution due to unforeseen limitations.

Scheduling Changes

Scheduling changes impact the approved project timeline, potentially requiring the team to crash the schedule or negotiate an extension for scheduled releases.

Budget Changes

Modifications of financial resources required, which in this context primarily relates to funds needed to upgrade cloud hosting tiers if user traffic exceeds to free limits.

Configuration Change Request (CCR)

A formal document drafted when feedback collected from Pet Owners (PO) or Service Providers (SP) within OpenProject necessitates a departure from the established baseline. All proposed system enhancement or problem resolution must be processed as a CCR.

Change Control Board

For the furlink capstone initiative, the Change Control Board (CCB) functions as the definitive approval authority for all proposed changes to the project's scope, timeline, budget, or multi-cloud infrastructure. Chaired by the Project Adviser and Co-Adviser, and including the Project Manager and Lead Developer, this collaborative group rigorously evaluates technical specifications and requested modifications against strict academic deadlines. The CCB ensures that any authorized changes are absolutely essential to delivering the core MVP and do not introduce scope creep that could jeopardize the final solution release.

Roles and Responsibilities

The following are the roles and responsibilities for all change management efforts related to the furlink Project:

Project Sponsor

- Review and approve any changes that directly impact on the project's budgets and schedule.
- Approve any requirements for additional financial backing.

Project Manager

- Receive and log all change requests from project stakeholders
- Conduct preliminary risk, cost, schedule, scope analysis of change prior to CCB
- Seek clarification from change requestors on any open issues or concerns
- Make documentation revisions/edits as necessary for all approved changes
- Participating in CCB

Project Team

- Submit all change requests on standard organizational change request forms
- Provide all applicable information and detail on change request forms
- Be prepared to address questions regarding any submitted change requests
- Provide feedback as necessary on impact of proposed changes

Project Adviser

- Ensure changes do not violate academic standards, course requirements, or timelines.
- Approve all major shifts in schedule baseline.
- Chair of Change Control Board

Lead Developer

- Draft formal CCRs for requested system departures.
- Authorize and create new feature branches in GitHub to isolate the development of approved changes, ensuring the stable production branch remains protected.

Lead Quality Assurance

- Verify implemented changes against the original user feedback.
- Ensure the enhancement or defect is fully resolved before authorizing the CCR to be marked as closed and integrated into the final release.

Change Control Process

The Change Control Process for the furlink project will follow the organizational standard change process for all projects. The project manager has overall responsibility for executing the change management process for each change request.

Change Classification and Criteria

To prevent bottlenecks, change requests for this project are classified into two categories:

Major Changes: Any modification that alters the project scope, timeline, core architecture, API, or budget. These must strictly follow the full Change Control Process.

Minor Changes: Low-risk, low-impact modifications that will not affect the sprint goal, database schema, or project baseline. These changes do not require a formal CCB assessment or a new Sprint planning alignment. The Project Manager and Lead

Developer can mutually approve these changes during the current Sprint Backlog as a task, if it does not jeopardize the current Sprint Backlog.

1. Identify the Need for a Change

The process begins when feedback or defect logs are collected from Pet Owners or Service Providers and documented within OpenProject. Any feedback necessitating a departure from the established software baseline is formulated and drafted as a formal Change Request by the assigned Full-Stack Developer.

2. Preliminary Impact Analysis

Once drafted, the Change Request is routed first to the Project Manager, who conducts a preliminary risk, cost, schedule, and scope analysis. During the analysis, the Project Manager determines if the change qualifies as a Major or a Minor Change. If it is a minor change, it bypasses the formal Change Control Board assessment, moving directly to implement during the current Sprint. If deemed viable, the Project Manager logs the request and formally submits it to the board.

3. Log Change Requests

The project manager will keep a log of all submitted Change Requests throughout the project lifecycle.

4. Obtain Decision on Change Request

The project team, acting collectively as the Configuration Control Board (CCB), will discuss the proposed change and decide whether it will be approved by evaluating it against two critical criteria: its alignment with digitizing the pet care market and its technical feasibility within the current tech stack.

5. Implement Change

If a change is approved by the CCB, the project manager will update and re-baseline project documentation as necessary and notify stakeholders. The Lead Developer will authorize a new Feature Branch in GitHub to isolate development. During implementation, any active CCRs or resulting blockers must be reported during daily stand-up meetings. Once built, the Lead Quality Assurance will verify the change against the original feedback before the CCR is closed and integrated into the stable term branch for final

6. Verification and Closure

Following implementation, the Quality Assurance lead must test and verify the code adjustments against the original user feedback to ensure the defect or enhancement is fully resolved. Only after this verification is complete is the Change Request marked as closed in OpenProject, allowing the Lead Full-Stack Developer to safely merge the code into the stable development branch for final release.

Monitoring and Controlling

Project Status Report

Project Status Summary

Scope	Schedule	Cost	Risks	Quality
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Percent Complete: 93%

Executive Overview:

The Furlink project is progressing completely on schedule according to our established timeline. The team spent the months of May and April finalizing database schemas and building the core booking workflows. The project is on track to enter our implementation phase during the Final Testing on July 6–7 and the official v1.0 Deployment on July 11, 2026.

The Schedule, Scope, Cost, and Quality constraints remain in Green. The project Risks parameter is held at Yellow because the upcoming July sprints has core database refinements, service provider training, and the v1.0 deployment within a single week, there is a tight deadline. Furthermore, the development of the Gemini 2.5 Flash AI Booking Assistant (July 20–30) introduces critical security and rate-limiting factors. The team is actively mitigating these risks by enforcing a strict 48-hour code freeze before integration windows and building secure, read-only Database Views to wall off sensitive revenue tables from prompt-injection.

Work Completed for Last Sprint

- Finalize Database Schema.
- Configure Supabase Auth and Row Level Security (RLS) policies.
- Develop frontend UI for Pet Owner and Service Provider profiles.
- Build the core calendar-based scheduling engine.
- Completed Role-Based Access Control setup.
- Completed the preliminary development for the core booking modules, preparing the system for upcoming testing.

Work Planned for Next Sprint

- Begin unit testing of the calendar-based scheduling engine to meet the target for zero double-booking.
- Implement the PayMongo transaction hooks and configure backend serverless webhooks to ensure booking slots only finalize upon a verified payment confirmation.
- Build secure, Read-Only SQL Database Views to act as a physical data barrier, keeping sensitive business revenue metrics hidden from upcoming AI prompts.
- Review interface layouts with the mobile-responsive constraints to meet the "Ease of Use" metrics before entering the July integration.

Open Issues

Issue Number	Description	Status
Issue #1	Need to restructure database access for the upcoming AI features to make sure the AI cannot query raw business tables.	Tech Lead optimizing the database before July integration.
Issue #2	Risk of missed testing windows due to late-stage code commits.	Project Manager implementing an unyielding 48-hour code freeze rule immediately preceding the July 6 testing phase.
Issue #3	Admin approval interface requires minor styling adjustments for absolute mobile responsiveness.	Assigned to frontend developers for next sprint.

Open Risks

ID	Risk Title	Description	Risk Response
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1	Low Provider Adoption	Targeted Makati service providers are hesitant to move away from manual booking.	Provide in-person onboarding and "concierge" profile setup services to lower barrier to entry.
2	AI API Cost Overrun	The intensive transactional token consumption driven by multi-turn dialogs within the AI Booking Assistant and massive data array parsing by the AI-Powered Data Analyst exceeds the core team's allocated fixed budget, causing premature deployment throttling, platform downtime, or financial operational deficits.	Implement programmatic server-side middleware rate limiting alongside robust token optimization, semantic compression arrays, and payload pruning rules within the Supabase Edge Functions.
3	AI Hallucination Risk	AI features generate incorrect schedules, pricing, or advice.	A hidden code guardrail checks all AI answers against the real database and forces the user to review a final summary screen before anything is officially booked.
4	Unauthorized Admin Access	Malicious actors bypassing security controls to gain full administrative privileges.	Implement multi-factor authentication (MFA) for admin panels and strict IP-whitelisting for internal tools.

5	Security Breach	Unauthorized access to sensitive user or payment data.	Implement HTTPS, industry-standard hashing, and strict Role-Based Access Control (RBAC).
6	UAT Failure	Pet owners find the interface confusing, leading to low "Ease of Use" ratings.	Conduct weekly UI reviews and iterative feedback sessions before the final August UAT.
7	Database Failure	Technical infrastructure and performance issues.	Set up point-in-time recovery (PITR) and automated daily database backups within your Supabase configurations.
8	Dashboard Data Processing	Critical crash of cloud database infrastructure, resulting in temporary loss of data.	Build a cache layer for dashboard data in Supabase; implement manual refresh buttons for providers.
9	Data Privacy Compliance Violation	Mishandling user consent or failing to honor user privacy rights, triggering legal penalties under the Philippine Data Privacy Act (R.A. 10173).	Design data registration screens to explicitly capture user consent via checkboxes, and build an automatic one-click "Delete Account" button so users can instantly wipe their records.
10	Loss of Customer Trust	Brand damage resulting from recurring technical issues, data breaches, or consistently poor user experience.	Publish real-time status pages; proactively dispatch compensation discounts to affected Makati pet owners.

11	Fraudulent Payment Shutdown	Fraudulent or suspicious transactions may trigger payment gateway security measures.	Deploy client-side transaction screening logic; hold booking validation until webhook confirms clear payment.
12	Payment Gateway Delay	Issues with PayMongo integration prevent automated payment link delivery.	Start development of the Payment Gateway early (July 22) and maintain manual link backup protocols.
13	High Traffic Issues	Server overload during peak booking seasons or hours, causing the application to significantly slow down or crash.	Configure Vercel auto-scaling thresholds to dynamically spin up serverless computing nodes during traffic spikes, implement a Supabase API caching layer to minimize heavy database reads, and set up a lightweight traffic-throttling middleware to queue incoming requests safely.
14	Mobile Device Compatibility Issues	The platform interface breaks, overlaps, or is rendering improperly on other devices.	Enforce mobile-first responsive CSS architecture and conduct automated cross-browser layout testing across standard iOS and Android screen profiles before every code deployment.
15	Incomplete System Updates	Deploying flawed or partially finished code to production, causing instability.	Establish a strict main-branch protection policy in GitHub requiring automated test passes and mandatory approval signatures from both QA and

			Release leads before code can be merged into production.
16	Third-Party Service Shutdown	Sudden discontinuation or prolonged outage of critical external platforms.	Abstract all third-party API dependencies into modular code wrappers, allowing the development team to quickly swap connections and migrate to an alternative provider with minimal code rewrites.
17	Down AI Image Generation	Outage of external AI tools preventing the creation of AI generated images.	Implement an automated fallback system in your code that temporarily routes image requests to a backup AI engine while the primary AI service is offline.

Deliverables and Milestones

Milestones	WBS	Planned	Forecasted	Actual	Status
Capstone Kickoff	2	1.2	March 7, 2026	March 7, 2026	Completed
Final testing		2.3	July 6 to July 7, 2026	July 6 to July 7, 2026	On schedule
Release to Service Provider (Training)		2.4	July 8, 2026	July 8, 2026	On schedule
Furlink v1.0 Deployment		2.5	July 11, 2026	July 11, 2026	On schedule
Enhancement of core booking modules		3.7	July 13-17, 2026	July 13-17, 2026	On schedule
Development of AI booking assistant		3.8	July 20-30, 2026	July 20-30, 2026	On schedule

System and Integration testing	3.9	July 31, 2026 – August 7, 2026	July 31, 2026 – August 7, 2026	July 31, 2026 – August 7, 2026	On schedule
Furlink v2. Release	3.10	August 10, 2026	August 10, 2026	August 10, 2026	On schedule
User acceptance testing & fixes	4.2	August 17, 2026 – August 21, 2026	August 17, 2026 – August 21, 2026	August 17, 2026 – August 21, 2026	On schedule
Solution release furlink v2.1	4.3	August 31, 2026	August 31, 2026	August 31, 2026	On schedule
Project completion	5.5	October 1, 2026	October 1, 2026	October 1, 2026	On schedule

Deliverables	WBS	Planned	Forecasted	Actual	Status
Create Project Charter	1.1.1	April 15, 2026	April 15, 2026	April 15, 2026	Completed
Create Business Case	1.1.2	April 21, 2026	April 21, 2026	April 21, 2026	Completed
Create Project Brief	1.2.3				Completed
Create Stakeholder Management Strategy	1.1.4	April 6, 2026	April 6, 2026	April 6, 2026	Completed
Create Stakeholder Analysis	1.1.5	April 6, 2026	April 6, 2026	April 6, 2026	Completed
Create Statement of Work	1.2.6	April 1, 2026	April 1, 2026	April 1, 2026	Completed
Plan Communication Management	2.1.1	April 6, 2026	April 6, 2026	April 6, 2026	Completed
Plan Scope Management	2.1.2	April 13, 2026	April 13, 2026	April 13, 2026	Completed
Create Work Breakdown Structure	2.1.3	May 11, 2026	May 11, 2026	May 11, 2026	Completed

Plan Schedule Management	2.1.4	April 13, 2026	April 13, 2026	April 13, 2026	Completed
Plan Quality Management	2.1.5	April 8, 2026	April 8, 2026	April 8, 2026	Completed
Plan Cost Management	2.1.6	April 2, 2026	April 2, 2026	April 2, 2026	Completed
Plan Human Resource	2.1.7	April 2, 2026	April 2, 2026	April 2, 2026	Completed
Create Agile Team Charter	6.1	April 6, 2026	April 6, 2026	April 6, 2026	Completed
Create System Design Document	7.1	April 6, 2026	April 6, 2026	April 6, 2026	Completed
Create Feasibility study	1.1.8	May 20, 2026	May 20, 2026	May 20, 2026	Completed
Create Business Model Canvas	1.1.7	April 29, 2026	April 29, 2026	April 29, 2026	Completed
Create Requirements Management Plan	2.1.9	May 8, 2026	May 8, 2026	May 8, 2026	Completed
Create Project Risk Management Plan	2.1.8	April 29, 2026	April 29, 2026	April 29, 2026	Completed
Create Risk register	2.1.10	May 1, 2026	May 1, 2026	May 1, 2026	Completed
Create change management plan	2.1.12	May 26, 2026	May 26, 2026	May 26, 2026	Completed
Create project monitoring and controlling project status report	4.1.1	May 4, 2026	May 4, 2026	May 4, 2026	Completed
Create transition out plan	5.1.1	May 6, 2026	May 6, 2026	May 6, 2026	Completed
Create Project acceptance	5.1.2	May 20, 2026	May 20, 2026	May 20, 2026	Completed
Create post project review	5.1.3	May 7, 2026	May 7, 2026	May 7, 2026	Completed

Create lessons learned	5.1.4	May 4, 2026	May 4, 2026	May 4, 2026	Completed
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Open Change Requests

The Furlink project uses a structured Change Control Process to evaluate all incoming Change Control Requests (CCRs) before updating the baseline plan. The process begins when a team member identifies a required modification and submits a formal CCR. Upon receipt, the Project Manager conducts a preliminary analysis evaluating the request's immediate impact on project risk, cost, schedule, and scope.

Following this initial process, the request is escalated to the Change Control Board (CCB) for formal governance. The CCB evaluates each proposed change against three critical criteria: its alignment with digitizing the local pet care market, its technical feasibility within our modern architecture, and whether it is essential to delivering the core Minimum Viable Product (MVP) within strict academic deadlines.

Once a change is officially approved by the CCB and signed off by the Project Adviser, it is integrated into the active sprint backlog, and its status is updated in the repository. To maintain team transparency and prevent timeline delays, any active CCRs, particularly those acting as project blockers are reported during daily stand-up meetings for immediate resolution. The following table format will be used for mapping out project records for any future open change requests:

Change request	Change request number	Request date	Current status
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Key Performance Indicators (KPI's)

At the completion of our initial phases, the furlink project continues to demonstrate exceptional financial and operational status. For both Sprint 1 and Sprint 2, our total combined baseline cost was projected at ₱135,300.00 where ₱89,650.00 is for Sprint 1 and ₱45,650.00 is for Sprint 2.

Combined actual expenditures were kept strictly to ₱123,000.00. This reflects ₱120,000.00 in total team labor across both phases and ₱3,000.00 in total co-working space allocations. Consequently, our entire cumulative 10% contingency reserve of ₱12,300.00 went completely unspent, resulting in a highly favorable Cost Variance. Operationally, all deliverables remain on-schedule.

Schedule - Project is On Schedule

- Schedule Variance (SV): ₱0.00
- Schedule Performance Index (SPI): 1.00

Cost - Project is Under Budget

Cost Variance (CV): ₱12,300.00

Closing

Transition Out Plan

Executive Summary

The purpose of this plan is to establish a high-level operational blueprint for the furlink Pet Grooming Platform. Unlike a traditional contract handover, this document ensures internal readiness and the preservation of technical "powers, duties, and functions" as the startup matures. It provides a roadmap for internal knowledge retention and the timeframe for transitioning from "Build Mode" to "Scale/Operations Mode" by October 1, 2026.

Transition Approach

Instead of scaling up staff, the startup will use this period to formalize roles.

- **Knowledge Consolidation:** Founders and Developers will transition from "sole knowledge holders" to "mentors" for new hires.
- **Assumption:** It is assumed that all documentation (API keys, logic flows, and vendor agreements) must be accessible to more than one person to prevent "single point of failure" risks.

Transition Team Organization

In a startup, team members often wear multiple hats. This chart defines the core responsibilities for long-term platform maintenance:

Title	Startup Role	Roles/Responsibilities
Product Owner	Founder	Oversee long-term strategy and roadmap; manages investor relations.

Technical Lead	Founder/Developer	Manages the Supabase backend, AI model fine-tuning, and documentation.
Ops & Growth Lead	Founder/Marketing	Manages service provider onboarding in Makati and daily business logic.
Release Lead	Founder/Developer	Manages the deployment pipeline; enforces code freezes; coordinates versioning and ensures the platform is live and stable.
QA Lead	Founder/Developer	Maintains the Traceability Matrix (RTM); oversees all testing phases (Unit, Integration, UAT); validates that Security Requirements are met.

Workforce Transition

For a startup, the workforce "transition" involves moving from a loose group of developers to a structured operations team. All current members will stay in place, but their focus will shift from Feature Development to System Reliability and Security Monitoring 15 days prior to the October 1, 2026, completion.

Work Execution During Transition

Throughout this phase, work continues according to the approved WBS. Responsibility for the AI Booking Assistant and payment integration is shared among the core team to ensure that the departure of any single member does not halt production.

Contract (SLA)

As a startup, these agreements are to protect platform integrity and establish institutional trust with our partner grooming service providers, furlink operates under a comprehensive, startup-scaled Service Level Agreement (SLA). This agreement explicitly details our commitment to a 99.99% core platform uptime while mapping clear operational boundaries regarding external supply-chain dependencies like Supabase, PayMongo, and Gemini API frameworks. Ultimately, this contract acts as a dual-purpose blueprint: it

protects furlink from liabilities tied to third-party infrastructural failures through automated manual fallbacks, while reassuring stakeholders and investors that our operational support mechanics are de-risked, formalized, and built for scalable quality assurance.

Link for furlink's service level agreement: [furlink Service Level Agreement.docx](#)

Asset & Intellectual Property (IP) Protection

Hardware & Training-Environment Access Management

- **Credential Security:** Access credentials for service providers' training accounts are generated securely and managed via our team vault. Under no circumstances are live administrative database keys or billing passwords exposed on training devices.

Intellectual Property (IP) Guardrails & Partner Data Segregation

- **Data Ownership:** While service providers retain the rights to their business operational metrics, the platform database architectures, user interface configurations, and the proprietary prompt configurations for the AI Booking Assistant remain 100% the intellectual property of furlink.
- **Anonymized Data Ingestion:** To train and fine-tune our AI features safely during provider validation phases, any operational feedback or booking text logs gathered from grooming partner workflows will be completely anonymized before system ingestion to protect user privacy.

Account Access Architecture & Multi-Store Management

- **Break-Glass Outage Protocols:** Master API overrides for the PayMongo and Gemini connections are securely distributed among the founding team. If a service provider encounters a system lockout or critical error during a live high-traffic booking window, any member of the founding team can execute immediate override diagnostics.

Knowledge Transfer

Knowledge transfer in a startup is the process of creating an Onboarding Guide for new service providers.

Documentation

Keep training guides live and updated in a shared workspace.

- Internal Wiki: Technical guides for the founding team to keep the system running.
- Partner Playbook: A non-technical guide used by the team to onboard service providers.

The "As-Built" Support Guide

Finished 15 days before launch, this guide allows *any* founder to give instant tech support to groomers when things act up:

- AI Troubleshooting: If a groomer says the AI Booking Assistant is misunderstanding clients, this guide shows how to quickly patch the prompt instructions.
- Payment Flow Maps: A simple checklist tracing how a customer's receipt upload updates the groomer's dashboard. This helps you instantly tell if a delay is a system bug or a bad user upload.

Core Team Cross-Training

The Technical Lead will run quick workshops so the entire team can handle emergency shop support:

- On-Site Diagnostics: Everyone will learn how to check if an app failure at a grooming shop is an actual system bug, a PayMongo/Gemini outage, or just bad shop Wi-Fi and tablet formatting.

Schedule

Schedule	Date
Release to Service Provider (Training)	July 8, 2026
furlink v1.0 Deployment	July 11, 2026
Enhancement of Core Booking Modules	July 13-17, 2026
Development of AI Booking Assistant	July 20-30, 2026
System and Integration Testing	July 31, 2026 – August 7, 2026
furlink v2.0 Release	August 10, 2026
User Acceptance Testing & Fixes	August 17, 2026 – August 21, 2026
Solution Release furlink v2.1	August 31, 2026
Project Completion	October 1, 2026

Final Operational Acceptance

The "Handover" is internally validated when the Founders sign off on the Operational Readiness Checklist. This confirms that all security protocols are active, all PRs are met, and the platform is ready for the pet owners and service providers.

Project Acceptance

This document establishes formal acceptance of the mid-project deliverables and current development baselines for the furlink platform, explicitly confirming that the project remains highly active and continuously ongoing. Rather than signaling a project closure or final termination, this acceptance serves as an institutional milestone to validate that the system has successfully met requirements defined for the current academic evaluation window. Because the development operations are still actively progressing, there will be no handover, turnover, or transfer of the platform to any external entity or third party. Absolute ownership, codebase control, and architectural governance of the project will remain permanently with the core project team. While the project advances through its active development lifecycle, the specific technical manuscripts and documentation required for institutional and academic completion will be compiled and submitted to the school.

Looking forward as the project continues to progress beyond the duration of the academic phase, the student team will remain fully intact, seamlessly shifting into their permanent roles as the long-term operations engine for furlink. In this ongoing operational capacity, the founders will directly handle all continuous platform maintenance, database optimization, infrastructure management, and market positioning. Concurrently, as the team drives the project towards its eventual academic phase completion, active field preparations are being conducted to ensure marketplace ecosystem readiness. This includes organizing and executing a specialized user navigation program, where the team conducts hands-on training for onboarding service providers guiding them to navigate the system. By executing this document, the team formally certifies the validity of the current project baseline states, authorizing the Project Manager to sustain active, ongoing development operations as the project team continuously refines the codebase and scaling furlink.

Post Project Review

Project Summary

The furlink platform was developed as an IT Capstone project to modernize the pet care industry by connecting dog and cat owners with service providers. The primary objective was to deploy a robust web application featuring service listings, a calendar-based booking system, and an innovative AI-powered haircut preview tool. Operating under strict academic deadlines, the project aimed to address visibility issues for local pet grooming businesses while providing a seamless scheduling experience for pet owners. Overall, the project successfully deployed its core MVP, allowing pilot service providers to onboard and manage bookings through a multi-cloud architecture utilizing Vercel and Supabase.

Project Team and Staffing

The project was executed by Team LogiTeh, operating within an Agile framework. The team successfully managed dual roles (academic and development) throughout the lifecycle:

Name	Project Role	Contact
Lalaine Andaya	Project Manager	lcandaya2@student.apc.edu.ph
Atasha Frances Gayle Doria	Lead Developer	asdoria@student.apc.edu.ph
Feliz Angelica Salting	Release Manager & Team Lead	fpsalting@student.acp.edu.ph
Michelle Reina Pineda	QA Lead	mbpineda@student.apc.edu.ph
Gonzalo Gumogda	Project Adviser	gonzalog@apc.edu.ph

Project Deliverables (Planned vs. Actual)

furlink Design

Planned Deliverable	Actual Deliverable	Summary
Design thinking research to conduct user interviews to build system requirements	Research conducted through interviews with service providers	This deliverable was completed as planned

Complete UI/UX wireframes and system designs	Completed system UI/UX wireframes and mockups, and system designs	This deliverable was completed as planned
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furlink Core MVP Development

Planned Deliverable	Actual Deliverable	Summary
Core platform modules including User Registration, Service Provider Listings, and the Calendar-Based Online Booking System	Core platform modules developed successfully with functional real-time calendar synchronization to prevent double bookings	This deliverable was completed as planned

furlink Testing and Quality Assurance

Planned Deliverable	Actual Deliverable	Summary
Comprehensive testing documentation covering Unit, Integration, System, and User Acceptance Testing (UAT) with early adopters	Comprehensive testing documentation tracking defect resolutions, automated test scripts, and UAT feedback from project adviser	UAT with early adopters were not implemented

furlink Final Project Deliverables

Planned Deliverable	Actual Deliverable	Summary
Service provider dashboard showing monthly sales, customer ratings, and market insights	Service provider dashboard showing monthly sales, customer ratings, and market insights	This deliverable was completed as planned

AI Pet Haircut Generator to preview grooming styles	AI Pet Haircut Generator deployed with an added standard booking fallback mechanism	Delivered with a strategic operational adjustment. A fallback bypass was added for sub-optimal photo uploads to prevent booking flow interruptions
Implement AI booking assistant and AI-powered data analyst	As scheduled	

Transition to Operations

Transitioning an academic Capstone project to a live, operational pilot environment requires ongoing, iterative effort rather than a one-time handoff. Team LogiTech ensures that the development team and the pilot service providers maintain effective communication throughout the project’s active lifecycle, ensuring continuity as the system evolves toward its final solution release this August. Additionally, the team continuously includes early-adopter businesses as Key Stakeholders throughout the ongoing development cycles.

As the platform matures, an immediate area for continuous improvement is conducting further iterative testing directly within the service providers' physical business environments (e.g., on-site inside the grooming salons), rather than limiting User Acceptance Testing to controlled academic or remote settings. Bringing the application directly into the salons allows service providers to gain deeper familiarity with the system while simultaneously surfacing real-world environmental constraints—such as salon Wi-Fi connectivity or specific mobile device limitations—ensuring a more robust and responsive final deployment.

Project Costs

The budgeted cost for the furlink platform was set at *** for three sprints. The following chart shows the project budget with actual costs compared to the planned cost.

Sprint	Budgeted Cost	Actual Cost	Comments
AY 25-26 TERM 3 SPRINT 1	89,650.00	37,965.00	
AY 25-26 TERM 3 SPRINT 2	45,650.00	39,357.00	

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Project Schedule

Milestones	Date	Comments
Capstone 2 Kickoff	March 7, 2026	Completed on time
Final Testing	July 6, 2026 – July 7, 2026	As scheduled
Release to Service Provider (Training)	July 8, 2026	As scheduled
furlink v1.0 Deployment	July 11, 2026	As scheduled
Enhancement of Core Booking Modules	July 13, 2026 – July 17, 2026	As scheduled
Development of AI Booking Assistant	June 20, 2026 – July 30, 2026	As scheduled
System and Integration Testing	July 31, 2026 – August 7, 2026	As scheduled
furlink v2.0 Release	August 10, 2026	As scheduled
User Acceptance Testing & Fixes	August 17, 2026 – August 21, 2026	As scheduled
Solution Release furlink v2.1	August 31, 2026	As scheduled
Project Completion	October 1, 2026	As scheduled

Recommendations

The furlink platform was an example of a carefully planned and successfully executed Capstone project for Team LogiTeh. As development and iterations continue, the following recommendations and lessons learned have been identified to guide the current lifecycle and assist future academic projects.

Recommendation #1: Offer direct on-site training and guided onboarding sessions with service providers to lower the barrier to adopt to a digital platform to mitigate low provider adoption risk.

Recommendation #2: Implement monitoring and buffer budgets for APIs to monitor token usage for the AI Booking Assistant and the AI-Powered Data Analyst to avoid cost overruns.

Recommendation #3: Integrate third-party payment gateways within the first version of deployment. To transition furlink from a booking facilitator to a fully self-contained e-commerce platform, integrating gateways like GCash or Maya will secure direct transaction and simplify user journeys.

Recommendation #4: Maximize cloud free-tier architectures to ensure that the project remains within the authorized budget.

Lessons Learned

Introduction

The purpose of the lessons learned document for the FurLink project is to progressively capture project insights in a formal document for use by future SoCIT (School of Computing and Information Technologies) PBL groups and student developers on similar projects. This document will detail the challenges that arise throughout the development cycle of the previous versions, how the team navigates them, and how they can be avoided in the future. Additionally, it highlights what goes well, such as early budget optimization strategies, so that other project managers may adopt these successful practices. Upon project completion, this document will be formally communicated to the Asia Pacific College for project advisers and will become a part of the academic and organizational archives.

Lesson Learned Approach

The lessons learned for the furLink project are continuously compiled from Agile sprint retrospectives, scheduled standups, GitHub/Vercel deployment logs, and OpenProject logs throughout the project lifecycle. Insights will also be actively drawn from both realized and unrealized risks monitored in the project risk register, specifically focusing on the integrations with the Gemini 2.5 Flash AI and the PayMongo payment gateway. Ultimately, the findings documented here will be categorized by project knowledge areas. These knowledge areas consist of risk management, cost management, scope management, stakeholder management, and quality management.

Lessons Learned from this Project

The following chart lists the lessons being learned for the furlink project. It is important to note that not only failures or shortcomings are included but successes as well.

Category	Issue Name	Problem/Success	Impact	Recommendation
Risk Management	External API Dependencies	Success: The team identified the risk of PayMongo webhook delays and Gemini AI rate limits early in the planning phase.	By allocating a dedicated, isolated testing sprint specifically for the payment gateway, the team avoided deployment blockers and maintained the August completion schedule.	Always allocate a dedicated buffer sprint for testing external third-party APIs before attempting to integrate them into the system workflow.
Scope & Security Management	AI Prompt Injection & Data Privacy	Problem: Initial AI integration tests revealed the Gemini Assistant could potentially be manipulated into accessing private business revenue tables.	Required an immediate architectural shift to implement Read-Only Database Views and strict Row-Level Security (RLS) to physically isolate the AI from sensitive data.	Never provide an LLM direct access to raw database tables. Always establish Read-Only Views and apply metadata filtering before integrating AI tools.
Quality Management	Pre-Deployment Code Stability	Problem: Late-stage code commits prior to integration testing caused serverless function cold-start delays.	Resulted in temporary performance drops on the Service Provider dashboard, requiring rapid Edge Function optimizations	Enforce a strict 48-hour "Code Freeze" prior to any major System/Integration testing window to ensure the testing environment is absolutely stable.

			just days before the v1.0 launch.	
Stakeholder Management	Digital Onboarding	Problem: Traditional grooming shops struggled with the initial self-service digital onboarding process.	There may user adoption during the MVP phase, as shop owners were unaccustomed to configuring digital profiles and liability waivers.	When digitizing traditional local businesses, do not rely purely on automated onboarding. Offer trainings to lower the barrier to entry.

Lessons Learned Knowledge Base / Database

The lessons learned for the Furlink Project will be contained in the organizational lessons learned knowledge base maintained by the APC-SoCIT PBL Repository. This information will be cataloged under the project's year (2026) and the technology stack for future reference. This information will be valuable for any future student project manager assigned to a web-based scheduling or AI-integrated capstone project.

Lessons Learned Knowledge Base / Database

The Furlink Project used several lessons learned from the past version of furlink:

- The decision to use Supabase (PostgreSQL) over Firebase was based on the note that handling complex relational data (like overlapping calendar appointments) was highly inefficient and expensive in NoSQL environments.
- The implementation of a strict 48-hour code freeze before Final Testing was adopted based on Agile ceremonies. Not doing so will cause the project to have severe delays due to untested hotfixes breaking the production build hours before their defense.

Process Improvement Recommendations

As indicated in the lessons learned chart above, integrating Large Language Models (LLMs) like Gemini into direct database queries introduces severe security risks (Prompt Injection and Data Leakage) that standard web applications do not face. Not only is this a

lesson learned for furlink, but the academic organization must ensure that all future project managers are aware of the need to secure their architectures against AI manipulation.

Therefore, it is recommended that APC-SoCIT standardize an "AI Security Checklist" for all future Capstone projects proposing LLM integrations. Operationally, future teams should mitigate adoption barriers by conducting on-site provider training and guided onboarding sessions. It is also recommended to implement real-time monitoring and buffer budgets for all AI services. Prior to development, teams must brief their project advisers on how they plan to implement Database Views and Restricted API Keys to make sure user data remains secure from AI related data leaks.

Agile

Agile Team Charter

Project Name	furlink
Vision	furlink's vision is to modernize and professionalize the pet grooming industry by providing a centralied digital platform that bridges the gap between pet owners and service providers.
Mission	Provide a hassle-free experience for pet owners and pet groomers through digitalized scheduling and innovative AI tools.
Primary Objectives	The system is designed to achieve four primary goals: 1. Ability to Customize Pet Haircut: Successfully develop and deploy an AI-powered haircut preview tool, integrated into the platform, by August 2026, ensuring it achieves a minimum of 70% image rendering accuracy, accurate identification of selected haircut suitability for the pet's current hair style and condition. 2. Reduce Scheduling Conflicts: Eliminate double booking by ensuring that each service provider does not experience booking incidents during the first three (3) months of the release by integrating an AI Booking Assistant. 3. Improve Visibility of Pet Service Providers: Improved the visibility of small to medium-sized pet care service providers by giving them a dedicated platform to showcase their services and directly connect pet owners. Onboard and verify one (1) service provider within the first three (3) months of release.

	4. Provide Business Insights for Service Providers: Enable pet care service providers to track monthly sales, customer feedback, and related insights by integrating a business analytics dashboard and AI-powered Data Analyst into the platform by August 2026.		
Success Criteria	• AI Haircut Preview: The AI tool must be ready by August 2026, hitting at least 70% accuracy when showing what a pet’s haircut will look like based on its current fur condition. • Zero Double-Bookings: Because of the AI Booking Assistant, no grooming shop should experience a single double-booking during the first 3 months after launch. • Service Provider Onboarding: Successfully verify, activate, and get at least 1 local grooming shop live on the app within 3 months of launching. • Active Storefronts: 100% of onboarded shops must have a working, public profile page so pet owners can find and contact them immediately. • Business Dashboard: Build a working sales and feedback tracking chart for service providers by August 2026. • AI Data Analyst: Deploy the AI Data Analyst by August 2026 so shop owners can automatically get summaries of their monthly performance. • Instant Payments: Ensure PayMongo payment links are generated instantly and without errors the moment a booking is confirmed. • Final App Sign-Off: Finish all user testing by August 31, 2026, with all critical bugs fixed and a user friendliness score of at least 3.5 out of 5 stars.		
Project Team			
Name	Role	Responsibility	Email
Lalainne Andaya	Project Manager	Project coordination, stakeholder communication	lcandaya2@student.apc.edu.ph
Atasha Frances Gayle Doria	Lead Full-Stack Developer	Architectural design, frontend/backend integration	asdoria@student.apc.edu.ph
Feliz Angelica Salting	Full-Stack Developer	Manage day-to-day development activities	fpsalting@student.apc.edu.ph

Michelle Reina Pineda	QA Lead	Testing procedures, bug management	mbpineda@student.apc.edu.ph
Rules of Behavior			
Contribution • Active Engagement: Every team member must commit to communicating clearly and actively, ensuring everyone shares the necessary updates required for project progress. • Feedback Integration: Everyone must actively respond to feedback given by the team, project advisers, or the steering committee. • Active Contribution: Every team member must contribute to the charter and sprint planning to maintain a sense of ownership. • Proactive Innovation: Every team member must take the initiative to bring realistic new ideas that push the boundaries of the platform's current capabilities. Transparency • All blockers must be reported during Daily Stand-ups to ensure the schedule remains on track. Accountability • Quality and Assurance: Every team member must provide results that exceed expectations, ensuring the work is accurate to minimize the need for re-work. • Active Accountability: Team members are responsible for moving their own OpenProject work packages to Closed after the work has been tested and verified Professionalism • All internal and external communication must follow the standards defined in the project's Communication Plan			
Team Communication			
As an Agile team, we prioritize interpersonal communication over formalized documentation. • We will conduct daily stand-up meetings via the PROJMAN group channel to discuss progress and immediate blockers. • We will make every effort to attend all scheduled meetings in person or online, in exception for members who feel unwell/sick. • The team will update tasks on OpenProject. • Meeting minutes must be provided within 24 hours after the meeting. • If a meeting is cancelled or additional meetings are required, the project team lead will send out notification as soon as possible. • All team members are expected to be on time for all meetings.			

Agile Team Values

I, as a member of **LogiTech** promise to do my very best to abide by the values of our group listed below. We have discussed and agreed to follow the values listed below.

I understand that I will evaluate myself and my peers based on the agreed upon values in an objective manner during the midterms and finals of the term.

Group Leader: (Feliz Angelica Salting, , March 16, 2026)

Members:



Lalainne Andaya March 16, 2026

Team Member



Atasha Frances Gayle Doria March 16, 2026

Team Member



Michelle Salting March 16, 2026

Team Member

Reina Pineda March 16, 2026



Feliz Angelica

Team Leader

Shared Values:

1. Communication

a. We communicate clearly and actively

- -1: Unresponsive during discussions and lack the needed information
- 0: Inadequate communication
- 1: Actively share updates that are needed for the progress.

b. We listen and respond to feedback

- -1: Dismisses feedback or ignored feedback from others.
- 0: Listens to others' feedback and responds appropriately.
- 1: Actively listens, asks questions, engages in discussion, and actively improves.

2. Ambitious

a. We take the initiative to come up with new ideas

- -1: Doesn't contribute any ideas to the team
- 0: Contributes standard ideas
- 1: Brings in new ideas that are realistic and pushes boundaries

b. Creating opportunities for the group

- -1: Does the work task poorly
- 0: Executes the assigned work task properly
- 1: Makes actionable plans for the group

3. Resilient

a. We adapt to change

- -1: Struggles to adjust to changes.
- 0: Adjusts to changes when necessary.
- 1: Quickly adjusts to changes and helps others adjust as well.

b. We bounce back from hindrance

- -1: Challenges significantly affect performance.
- 0: Recovers from hindrance and continues working towards the goals.
- 1: Takes challenges as a learning opportunity and works towards improving future outcomes.

4. Reliability

a. We consistently contribute meaningful progress towards the goal that we're working through

- -1: No significant contribution to the group in achieving the goal.
- 0: Slightly significant contribution to the group in achieving the goal.

- 1: Highly contributes to the group towards achieving the goal.
- b. We provide accurate results
- -1: Results are inaccurate and will take time to re-do work.
 - 0: Results are accurate with little to no re-work to be done.
 - 1: Results provided were beyond expectations.
- c. We meet deadlines
- -1: Contributions or outputs were made late.
 - 0: Contributions or outputs are on time.
 - 1: Contributions or outputs are made before the deadline.

Project Documents

System Design Document

Introduction

This document outlines the technical architecture for furlink. furlink is designed to replace a manual booking process with an integrated digital ecosystem. The system centralizes pet registration, scheduling, and payment processing into a single platform accessible via the web.

Purpose

The purpose of this document is to ensure the furlink design meets established project requirements for real-time booking and business management. It provides a detailed description of the system architecture, software modules, and security protocols to ensure organizational goals for efficient pet care management are achieved.

System Overview

This section describes the core architectural goals and operational approaches of furlink. Designed to replace manual, fragmented coordination methods that frequently lead to scheduling conflicts and inconsistent record management, furlink standardizes the

digital marketplace experience to optimize scheduling efficiency, data visibility, and workflow transparency for both pet owners and service providers.

The system is designed to achieve four primary goals:

- **Ability to Customize Pet Haircut:** Successfully develop and deploy an AI-powered haircut preview tool, integrated into the platform, by August 2026, ensuring it achieves a minimum of 70% image rendering accuracy, accurate identification of selected haircut suitability for the pet's current hair style and condition.
- **Reduce Scheduling Conflicts:** Eliminate double booking by ensuring that each service provider does not experience booking incidents during the first three (3) months of the release by integrating an AI Booking Assistant.
- **Improve Visibility of Pet Service Providers:** Improve the visibility of small to medium-sized pet care service providers by giving them a dedicated platform to showcase their services and directly connect pet owners. Onboard and verify one (1) service provider within the first three (3) months of release.
- **Provide Business Insights for Service Providers:** Enable pet care service providers to track monthly sales, customer feedback, and related insights by integrating a business analytics dashboard and AI-powered Data Analyst into the platform by August 2026.

The furlink architecture utilizes a centralized, shared cloud database engine to ensure all operational datasets remain fully synchronized and consistent across modules in real time. The web presentation layer is engineered using React.js hosted on Vercel, with Supabase leveraging a cloud-native PostgreSQL instance to serve as the secure backend-as-a-service (BaaS) and database repository. Furthermore, PayMongo is securely integrated via API endpoints as the official external payment clearing gateway, while Supabase Edge Functions orchestrate all serverless background processing, including distributed AI model

execution, Gmail SMTP transactional notifications, and automated payment webhook handshakes.

Design Constraints

The engineering and implementation of the furlink platform are bound by several critical technical, operational, and regulatory constraints. These boundaries originate outside the immediate application scope but must be structurally factored into the system architecture to ensure predictable financial runtimes, platform stability, and compliance.

- **Hardware and Technology Constraints**
 - **Continuous Cloud Dependency:** The platform functions as a purely cloud-native, serverless architecture. It possesses a non-negotiable requirement for an active, low-latency internet connection to execute core operations, including live marketplace listings, asynchronous slot scheduling, data schema synchronization, and secure transaction workflows.
 - **Serverless Compute Cold Starts:** Backend functions deployed via cloud layers are bound by cold start constraints. Periods of platform inactivity can introduce a transient processing latency of 1 to 3 seconds during initial invocation, impacting immediate API responsiveness, real-time message routing, and background webhook processing.
 - **AI Rate Limiting:** To mitigate the high risk of AI API Cost Overrun and protect the team's budget allocation, strict programmatic rate limits are enforced at the backend middleware layer. While concurrent traffic interacting with the AI Booking Assistant, AI Pet Haircut Generator, or AI-Powered Data Analyst could cause transaction queuing or delayed execution if user caps are met, this throttling design constraint is a deliberate financial guardrail.
 - **Generative Image Latency and Dual-Provider Fallback:** The AI Pet Haircut Generator is constrained by heavy GPU compute cycles required to render high-fidelity graphics. To balance latency, computational availability, and resource expenditure, the system employs a tiered configuration strategy

utilizing Hugging Face Serverless Inference as the primary image engine. If Hugging Face encounters transient request timeouts or configured image generation fallback, meaning users may experience a variable rendering latency of several seconds depending on the active provider.

- **API Service Availability:** The text-based intelligent modules—specifically the AI Booking Assistant and the AI-powered Data Analyst—rely exclusively on the Google Gemini 2.5 Flash API with no secondary fallback model engineered into the pipeline. This single-model dependency is justified by two architectural realities: first, real-world telemetry confirms that Google’s enterprise infrastructure maintains an exceptional 99.9% operational availability, rendering its cumulative 8 hours and 45 minutes annual distributed downtime an acceptable operational risk; second, because conversational booking guidance and analytical report parsing represent non-critical enhancements rather than core utilities, any transient Gemini outage will not disrupt the platform’s primary database booking engine or PayMongo clearing workflows. If an outage occurs, the frontend gracefully degrades to standard, rule-based database pathing without compromising application uptime.
- **Software and System Integration**
 - **Two-Tiered Booking Finalization and Webhook Coupling:** The finalization of a pet grooming appointment is strictly decoupled into a two-tiered state machine. The system design is constrained such that receiving a successful payment confirmation signal from the external PayMongo webhook engine does not automatically finalize a booking; it merely transitions the database record into a secured, paid request state. Absolute finalization and database state persistence of an appointment are legally and programmatically coupled with a secondary, manual review and acceptance signal triggered by the designated Service Provider. Any third-party webhook drop or provider

connectivity lag directly holds the transaction in an intermediate reservation queue.

- **Role-Based Access Control:** The application layout enforces strict data isolation across three distinct user dimensions: Pet Owners, Service Providers, and Application Administrators. This structural constraint requires separate permission profiles and Row-Level Security (RLS) tokens to be securely evaluated across all serverless endpoints to prevent lateral data privilege escalation.
- **AI Context Window Dependency:** The continuous contextual awareness of the AI Booking Assistant and the AI-Powered Data Analyst is constrained by the mathematical token boundaries of the Gemini context window. Extended chat sessions that accumulate history beyond this memory parameter require the system to drop oldest messages or prompt the user to re-input parameters to prevent processing errors or conversational context loss.
- **Standards and Legal Rules**
 - **Data Privacy Act (R.A. 10173) Compliance:** To comply with the Philippine Data Privacy Act of 2012, all stored personal, pet health, and commercial transaction records must adhere to strict protection, minimization, and right-to-erasure mandates. The AI Booking Assistant is programmatically restricted from accessing any user data outside of what the user has explicitly typed or stored in their active session. Similarly, the AI-Powered Data Analyst is sandboxed to compile metrics strictly from aggregated, anonymized business data belonging to the authenticated service provider, completely blocking cross-tenant data leaks.
 - **Payment Architecture Decoupling:** To align with global digital payment security standards, the platform is strictly prohibited from capturing, processing, or caching financial card data or bank credentials within its internal database. The AI Booking Assistant's capabilities are legally restricted to generating encrypted PayMongo checkout strings, completely transferring the

transactional data-entry phase onto PayMongo's isolated, compliant external environment.

- **Pet Care Operational Workflows:** The system must strictly enforce standard veterinary and pet care compliance standards. This includes mandatory verification parameters for uploaded vaccination histories, age boundaries for specific treatments, and illness disclosure fields before a grooming appointment request can be forwarded to a provider.
- **Transactional Audit Accountability:** A permanent, immutable audit trail must be recorded automatically for every commercial booking transaction, escrowed payment state, provider acceptance decision, and administrative change request. This logging constraint ensures total platform transparency and operational accountability in line with standard consumer service industry practices.

Roles and Responsibilities

The table below defines furlink System Design roles and responsibilities and serves as the list of points of contact for issues and concerns:

Name	Role	Email
Lalaine Andaya	Project Manager	lcandaya2@student.apc.edu.ph
Atasha Frances Gayle Doria	Lead Full-Stack Developer	asdoria@student.apc.edu.ph
Michelle Reina Pineda	Quality Assurance	mbpineda@student.apc.edu.ph
Feliz Angelica Salting	Full-Stack Developer	fpsalting@student.apc.edu.ph

Project References

furlink system is design is guided by a combination of regional market analyses, localized consumer data, digital commerce standards, and comparative platform case

studies. These references serve as the baseline requirements for resolving inefficiencies and visibility limitations identified within the local pet care ecosystem. The following references inform the platform’s architectural, data management, and user experience design decisions:

Philippine Pet Care Market Report (2025): Documents the expansion of the domestic pet care sector to a \$1.2 billion USD valuation, detailing the 22% annual growth rate in grooming services that justifies the technical scalability of the furlink database repository.

Vero ASEAN Consumer Behavior Survey (2024): Established that Filipino pet owners allocate an average of 15% of their household budgets to pet-related expenses, serving as the basis for the user interface metrics and target audience definitions.

Metro Manila Pet Service Provider Logistics Survey: Outlines the operational benchmarks of local grooming businesses, detailing how a 70% reliance on informal messaging channels results in a 33% double-booking rate, which directly defines the core system requirements for furlink’s automated calendar synchronization modules.

Platform UX Case Studies (Grab, Agoda, and Foodpanda): Provides the architectural blueprints for modular platform design, geo-location searching, and seamless multi-channel navigation layouts.

Conversational Commerce Framework (Salesforce, Infobip, and Bloomreach Standards): Establishes the technical logic, intent-parsing parameters, and dialogue flow methodologies required to construct the real-time interaction capabilities of the AI Booking Assistant.

Comparative B2B/B2C Platform Reviews (PetBacker, Rover, and DaySmart): Identifies functional gaps in existing market options—such as a lack of upfront pricing transparency and localized availability—which directly guide furlink’s custom data flows for registration, instant quotation generation, and salon listing modules.

System Architecture

The architecture necessary to achieve the system design for furlink consists of following components:

Hardware

The furlink ecosystem is engineered as a fully cloud-based environment and rejects any dependency on physical or on-premises server hardware. Computing allocations, routing infrastructure, and file storage architectures are managed exclusively through specialized third-party cloud platforms.

- Vercel: Hosts and deploys the frontend web application with automated builds and continuous deployment connected to the GitHub Repository. It also serves as the authoritative domain registry and DNS management hub, anchoring all client traffic to the official system domain.
- Supabase: Manages the relational PostgreSQL database instances, handles core computing triggers, provisions secure cloud user authentication servers, and isolates encrypted binary file storage buckets.
- User Devices: The architecture restricts hardware dependencies entirely to the end-user side. The web platform is universally accessible via any modern consumer device running a standard, browser-capable application layer, completely removing the need for proprietary local hardware.

Software

Frontend

- React.js: Functions as the core foundational framework for client-side user interface construction, state management, and real-time interface updates.
- Vite: Utilized as the high-speed bundler and development server, executing highly optimized production code compilation and fast refresh modules to maximize development lifecycle speeds.

- Chart.js Library: Integrated directly into the provider interface to drive the interactive Business Dashboard, executing high-performance rendering of client-side canvas data visualizations to map sales trajectories, booking velocities, and provider revenue matrices.
- Vercel AI SDK: Implemented to establish streaming communication pipelines between the React presentation view and downstream artificial intelligence nodes, optimizing multi-turn chat performance, orchestrating local user prompt states, and receiving structured payload responses.

Backend

- Supabase Auth with JSON Web Token (JWT): Manages user session security by issuing cryptographically signed JWT keys upon successful login, verifying identities across subsequent transaction requests.
- Supabase Row Level Security (RLS): Enforces strict database-level access boundaries by automatically evaluating JWT claims against predefined SQL access policies, ensuring users only interact with their explicitly authorized data rows.
- Supabase Edge Functions: Handles serverless background logic including webhooks, AI orchestration, and automated triggers. The Gemini API key is stored securely as a Supabase secret and is never exposed to the client.
- Gmail SMTP: Delivers transactional email notifications triggered by Edge Functions.

Payments

- PayMongo: Payment gateway integrated via API for booking payment processing and webhook-based confirmation.

Artificial Intelligence

- Google Gemini 2.5 Flash: Powers both the conversational AI Booking Assistant and the analytical AI-Powered Data Analyst modules. This engine was selected for its high-speed transactional speed, cost-efficient token mapping, and extensive context window, which allows it to naturally retain comprehensive pet profiles and conversation histories across multi-turn booking dialogs or massive provider sales datasets.
- Vercel AI SDK with @ai-sdk/google provider: Orchestrates communication between the React frontend and the Gemini model, enabling tool calling for database queries and structured data extraction for form autofill, as well as a dynamic chart and report generation for the AI-Powered Data Analyst.
- Image Generation API: Powers the backend execution for the AI Pet Haircut Generator module, isolating intensive graphic rendering away from client-side resources. The architecture implements a tiered dual-provider configuration strategy handled via Supabase Edge Functions. The system establishes Hugging Face Serverless Inference as the primary graphic processing node to generate visual pet style previews from uploaded user photographs. If the Hugging Face API triggers a transient rate limit, timeout, or quota restriction, the backend framework executes an automated failover handshake to route the image payload to Pollinations.ai as the backup generation container, ensuring high application uptime while shielding private vendor tokens from exposure.

Version Control, Governance, and Repository Frameworks

- GitHub Repository and CI/CD Pipelines: Functions as the absolute single source of truth for the furlink codebase. It enforces strict version control boundaries, protects development branches, and forms the

core of the continuous integration and continuous deployment (CI/CD) infrastructure by triggering automated, production-ready builds on Vercel the moment an approved code modification is safely merged into the main development line.

- GitHub Codespaces: Serves as the team's unified, cloud-hosted virtual development infrastructure, providing containerized developer environments to ensure absolute runtime consistency across parallel programming sprints.

Software Modules

The following modules operate through the shared Supabase backend:

- Authentication and Access Control Module
- User Profile Management Module
- Pet Profile Module
- Service Provider Onboarding Module
- Platform Admin Oversight and Moderation Module
- Transaction and Payment Module
- Provider Operations Module
- Cancellation and Rescheduling Module
- Business Intelligence Dashboard Module
- AI Booking Assistant Module
- AI Pet Haircut Generator Module
- AI-Powered Data Analyst Module
- Discovery and Filtering Module

Database Design

The database layer of furlink is provisioned and hosted natively on Supabase, leveraging a cloud-based, high-performance PostgreSQL relational database engine. Functioning as the absolute single source of truth and centralized data repository for the entire platform, this layer guarantees that data inputs processed through any interface entry point are immediately consistent, normalized, and accessible across parallel software modules. To optimize data distribution and security, all database tables implement universally unique identifiers (UUIDs) as primary keys, eliminating sequential enumeration vulnerabilities and ensuring collision-free record generation. Furthermore, the entire schema architecture is strictly governed by PostgreSQL Row-Level Security (RLS) policies that enforce data access boundaries directly at the engine layer. These cryptographically checked RLS parameters validate user JSON Web Tokens (JWT) upon every transaction, ensuring that authenticated Pet Owners, Service Providers, and Application Administrators can execute data manipulation commands strictly within their pre-authorized boundaries. Because this data governance applies independently of the presentation view, any unauthorized structural operations or cross-tenant queries are rejected at the database level even if a malicious request successfully bypasses front-end application logic.

The database is organized into seven logical groups corresponding to the platform's core functional areas:

- Identity and Access
- Pet Management
- Service Provider
- Services and Pricing
- Booking Flow
- Post-Service
- Notifications

Each group contains tables that are directly linked through foreign key relationships to maintain referential integrity across the system. Several automated behaviors are enforced at the database level through PostgreSQL triggers and Edge Functions,

including real-time email and in-app notifications, automatic booking decline after 24 hours of no service provider response, and automatic booking completion 3 days after the scheduled data to unlock the review window. The AI Booking Assistant also interacts with the database through secured Edge Functions, querying the tables for availability and information searches and writing when auto-filling and confirming bookings.

Hardware and Software Detailed Design

Hardware

furlink does not utilize or maintain any physical or on-premises server hardware, as the entire core infrastructure is deployed across third-party cloud service providers and managed serverless environments as documented in the System Architecture section.

Software

furlink's software architecture is organized into highly cohesive modules designed to isolate distinct boundaries of platform functionality. The system layout was engineered around the specific operational needs of pet owners, service providers, and application administrators, and was compartmentalized into decoupled modules to allow independent development, targeted testing, and parallel maintenance. All software modules communicate asynchronously through the shared Supabase backend engine and are supported by standalone architectural diagrams for detailed technical reference:

- **Authentication and Access Control Module:** Manages registration, login, and role assignment for three user types through Supabase Authentication and JWT, with permissions enforced through RBAC and RLS.
- **User Profile Management Module:** Allows users to update personal details, change passwords, view their role, and deactivate their account via soft-delete mechanism.

- **Pet Profile Module:** Enables pet owners to create and maintain permanent pet profiles that can be referenced during the booking process to reduce data entry.
- **Service Provider Onboarding Module:** An application workflow collecting business credentials, service offerings, pricing, staff, operating hours, and supporting documents for application admin review.
- **Platform Admin Oversight and Moderation Module:** Provides the application administrator with tools to approve or reject service provider applications and manage user warnings or suspensions.
- **Transaction and Payment Module:** Initiates a PayMongo checkout session upon booking, redirects the user to PayMongo's hosted page, and finalizes the booking only upon receiving a verified webhook confirmation. furlink does not handle raw card details at any point.
- **Provider Operations Module:** Gives service providers a real-time interface to accept or decline booking requests, with status changes reflected instantly on the pet owner's appointment page and booking history.
- **Cancellation and Rescheduling Module:** Handles cancellations, rescheduling requests, and post-service review submission, with review eligibility automatically unlocked three days after the scheduled booking date.
- **Business Intelligence Dashboard Module:** Presents service providers with visual summaries of business performance, sales performance, and customer insights.
- **AI Booking Assistant Module:** Implements a natural language interface powered by the Google Gemini 2.5 Flash API and orchestrated via the Vercel AI SDK to automate and simplify client scheduling workflows through three core capabilities:
 - Knowledge Retrieval allows the conversational assistant to scan active marketplace data—including provider service listings, real-time availability matrices, and general platform FAQs—by triggering

secured Supabase Edge Functions that execute read-only database queries and summarize the raw outputs into plain text for the user.

- Automated Data Entry interprets unstructured natural language chat streams to extract structured parameters (such as pet names, grooming selections, and date), automatically populating the hidden Pet Registration and Booking Forms for immediate user verification and manual confirmation.
- Payment Orchestration triggers a server-side Edge Function calling the external PayMongo API to generate a unique checkout link once the client confirms their data parameters. The assistant injects this secure URL directly into the chat stream for user completion, allowing the underlying database row to be safely updated only after a PayMongo webhook confirms successful clearing.
- AI Pet Haircut Generator Module: Available to pet owners during the booking pipeline if their selected service includes a styling haircut option. The component detects the pet's verified type and breed to return an explicit menu of standard haircut options. The user can select a style and optionally upload a photograph of their pet to receive an AI-generated preview illustrating how the styling configuration will look on their specific pet. This asset serves as an immutable visual reference linked directly to the booking record to optimize provider grooming accuracy. The generation logic is handled server-side via a Supabase Edge Function to protect the private access keys. The uploaded user photograph and the resulting generated preview are stored securely within the database rows as `ai_reference_url` and `ai_generated_url` respectively. To ensure maximum uptime, a dual-provider redundancy layer is engineered into the Edge Functions: the system primarily executes the request via a high-performance model hosted on a Hugging Face Inference Endpoint; if the primary endpoint returns an error, timeout, or quota restriction, an automated failover instantly reroutes the payload to

Pollinations.ai as the backup generation container, eliminating user friction during peak periods or third-party outages.

- **AI-Powered Data Analyst Module:** Integrated exclusively within the provider-side Business Intelligence Dashboard. The module introduces an interactive conversational interface powered by the Google Gemini 2.5 Flash API that allows service providers to execute natural language queries against their local operational metrics and receive analytical insights, custom data arrays, and structured performance trends in return. The module calls secured Supabase Edge Functions that are strictly sandboxed via custom SQL views, ensuring that each provider can only access and analyze records belonging to their own authenticated account, completely blocking cross-tenant data visibility. This module extends the existing Business Intelligence Dashboard Module by adding a dynamic, conversational layer on top of the static visual summaries already available.
- **Discovery and Filtering Module:** Allows pet owners to search and filter service providers by location, star rating, and price range.

System Security and Integrity Controls

furlink incorporates multiple layers of security and integrity controls through a multi-tiered approach covering authentication, authorization, financial tokenization, payment security, storage, data integrity, and operational continuity.

- **Authentication and Authorization:** Core access control is enforced through Supabase Auth utilizing cryptographically signed JSON Web Tokens (JWT). Upon successful identification, tokens are securely issued to the client browser, scoped to the individual's specific system role, and set to automatically expire after a defined session period. Role-Based Access Control (RBAC) rules restrict Pet Owners, Service Providers, and Administrators to their explicitly authorized platform modules. Simultaneously, Supabase Row-Level Security (RLS) policies

evaluate these JWT parameters directly at the database engine layer to block unauthorized operations, ensuring that suspended or deactivated user accounts are rejected regardless of frontend presentation states or token generation anomalies.

- **Input Validation:** Validation rules are enforced across all forms to prevent malformed or malicious data from entering the system. These rules cover required fields, data types, character limits, and format constraints, and are documented in the project's test cases.
- **Payment Security:** To guarantee absolute transactional compliance with global payment cards industry standards, furlink completely decouples its infrastructure from financial data handling by utilizing PayMongo's secure, hosted checkout ecosystem. Raw credit card numbers, banking credentials, and digital wallet keys are never collected, processed, or cached within the platform's database. Financial finalization remains strictly frozen until an automated, cryptographically signed webhook confirmation signal is transmitted from PayMongo's PCI-compliant servers to the Supabase Edge Functions. Under this framework, the conversational AI Booking Assistant is programmatically restricted from accessing or storing billing details, functioning solely as an encoder to render external transaction links.
- **AI Security:** Artificial intelligence orchestration keys are securely isolated away from public application code. The Google Gemini enterprise token, the Hugging Face Serverless Inference key, and all alternative Pollinations.ai access vectors are stored exclusively as encrypted Supabase environment secrets, executing entirely within server-side Edge Functions. To align with the Data Privacy Act of 2012, system prompt boundaries strictly blind the AI Booking Assistant from reading user logs outside of the active session context. Concurrently, the AI-Powered Data Analyst is securely sandboxed via isolated SQL views, ensuring its text-generation routines can only query aggregated historical telemetry belonging

explicitly to the authenticated provider account, neutralizing lateral cross-tenant data leaks.

- **Storage and Media Security:** Pet media files including vaccine records, illness documents, pet reference photos uploaded for the AI Pet Haircut Generator, and AI-generated haircut preview images are stored in Supabase Storage and are planned for migration to private bucket access, where files will only be accessible through time-limited signed URLs tied to authenticated user sessions.
- **Account Integrity:** User deactivation requests utilize a structured soft-delete mechanism that immediately flags the account status row as inactive and revokes database read-write tokens while preserving historical logs for transaction tracing and compliance audits. Complete account purging and irreversible database row deletions are executed strictly upon the submission of a formal data erasure request, maintaining complete alignment with the mandates of the Philippine Data Privacy Act of 2012 (R.A. 10173).
- **Audit Trail:** Login events are recorded in the `user_sessions` table, and database triggers across key tables log all significant data modifications, providing a traceable record of system activity for administrator review.
- **Operational Continuity:** Platform availability is secured through an infrastructure layout that balances vendor redundancy with local failover protocols. The core presentation layer and database schemas inherit the high-availability clustering, automated daily snapshot distributions, and routine background backup configurations built directly into Vercel and Supabase cloud topologies. For distributed AI features, continuity is split by operational impact: the single-model dependency on Google Gemini for conversational modules represents an accepted project risk backed by its 99.9% uptime, utilizing a frontend graceful degradation protocol to route users to standard database menus during rare outages. For the intensive graphic rendering pipeline, continuity is maintained through an active multi-model failover mechanism inside the Edge Functions that

instantly reroutes traffic from a timed-out Hugging Face inference gateway to a backup Pollinations.ai container, minimizing user friction.